Globalized consumption offsets efficiency gains across planetary boundaries

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Despite decades of ambitious environmental policy, German consumption has persistently exceeded planetary-boundary thresholds in six of seven impact categories since 1995, with little improvement over time. We combine environmentally extended multi-regional input–output tables with life-cycle impact assessment to develop a consumption-based sustainability indicator across seven impact categories, benchmarked against national planetary-boundary thresholds. Decomposing impacts into scale and impact intensity per euro of production by origin, we find that imports embody more impact per euro than domestic production in every category. A rising import share thus slows aggregate efficiency gains even as intensities fall in both origins, and the growing scale of imports largely offsets them. Had the import share remained at its 1995 level, cumulative impact growth from 1995 to 2022 would have been 12.9 to 96.8 percentage points lower across categories. We show that the same mechanism operates across comparable open, high-income economies.

Keywords: Planetary boundaries, Environmental footprints, Global value chains, Trade and environment

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Earth has crossed seven of nine planetary boundaries (Caesar et al., 2026), pushing critical Earth-system processes beyond the safe operating space that has underpinned human civilizations for millennia (Rockström et al., 2009; Steffen et al., 2015; Richardson et al., 2023). Halting this trajectory requires understanding not only whether environmental pressures are declining in absolute terms, but also where they originate. For an open economy, this means asking how much of the pressure behind its consumption—the basis of its standard of living—is generated abroad rather than at home. These questions are especially pressing for high-income economies, which combine high consumption levels with deep integration into global production networks. Yet existing policy mainly (i) relies on territorial production-based indicators, which count exports while excluding the imports that domestic consumption draws in, (ii) focuses on a narrow set of environmental dimensions, or (iii) sets targets against relative rather than absolute limits—each of which can give a misleading picture of progress.

We address these gaps by developing and applying a comprehensive, consumption-based sustainability indicator for Germany, in which consumption comprises all final demand except exports. Germany combines two characteristics that are highly relevant for such an assessment. On the one hand, its long-standing policy ambition and relatively stringent environmental regulation provide conditions under which efficiency improvements in production are plausibly substantial (Kruse et al., 2022; Erbach, 2024). On the other hand, its deep integration into global value chains exposes domestic consumption to international production processes and enables trade-related displacement of environmental pressures. These features make Germany a suitable case for examining whether efficiency gains observed under ambitious policy conditions translate into absolute sustainability once consumption patterns and international supply chains are fully taken into account.

Combining environmentally extended multi-regional input–output (EE-MRIO) tables with life-cycle impact assessment (LCIA) methods, we quantify environmental pressures across seven impact categories—climate change, particulate matter, photochemical ozone formation, land use, water use, fossil resource use, and mineral resource use—and evaluate them against national shares of planetary-boundary-consistent thresholds over the period 1995–2022. To shed light on the drivers behind environmental impact, we disaggregate impacts by geographic origin into a part generated abroad and a part generated in Germany. We further decompose

each of these components into the scale of production and a measure of impact intensity per euro of production for final demand in Germany. We deflate both measures with origin-specific price indices, so that their changes over time are not driven by price changes.

Our central finding is that German consumption has continuously exceeded national planetary-boundary thresholds in six of seven categories, with little improvement over time. Domestic production shows absolute decoupling in six of seven categories—aggregate pressure falling even as the economy grows—but once imports and upstream supply chains are counted, this decoupling weakens markedly or reverses. Two import-linked mechanisms drive this pattern. First, efficiency gains are outweighed by consumption growth fueled by rising imports. Second, a growing import share, embodying more impact per euro than domestic production, dampens aggregate efficiency improvements. To gauge the role of this shift, we recompute each consumption-based series under an accounting counterfactual holding the share of imports in the value of production for German final demand at its 1995 level, ceteris paribus: the rising share raised cumulative impact growth between 1995 and 2022 by 12.9 (climate change) to 96.8 (water use) percentage points. This mechanism is not a German idiosyncrasy. Replicating the analysis across the economies covered individually in EXIOBASE, we find that in every comparable open, high-income economy foreign production is more impact-intensive than domestic production in nearly every category-year, and that the rising import share raised impact growth in every impact category; boundary transgression itself is broadest and most persistent in the wealthiest economies.

Our paper contributes to several interrelated strands of the literature. First, it builds on the planetary-boundaries framework (Rockström et al., 2009; Steffen et al., 2015; Richardson et al., 2023), the growing body of work that translates planetary boundaries into LCIA-compatible indicators suitable for economic accounting (Bjørn et al., 2017; Ryberg et al., 2018; Sala et al., 2020), and the literature on safe and just operating spaces at the national level (Raworth, 2017; O'Neill et al., 2018; Fanning et al., 2022), whose applications typically focus on cross-sectional comparisons or short time horizons (Sanyé-Mengual and Sala, 2023). We extend this literature in three directions: (i) by providing a long-run assessment spanning nearly three decades; (ii) by combining absolute planetary-boundary thresholds with a structural decomposition that separates the scale of production from impact intensities per euro of production, disaggregated

by domestic and foreign production, which allows us to go from documenting boundary transgressions to explaining why they persist—a link that decomposition studies, largely confined to carbon and energy footprints (Hubacek et al., 2021; Baiocchi and Minx, 2010; Jiang et al., 2021; Dietzenbacher et al., 2020), have not yet made over a long time horizon; and (iii) by replicating the decomposition across countries, which shows that the mechanism recurs across comparable open, high-income economies. Second, we contribute to the literature on consumption-based environmental accounting and the role of international trade in shaping national environmental footprints (Wiedmann, 2009; Peters and Hertwich, 2008; Hertwich and Peters, 2009; Wiedmann and Lenzen, 2018; Peters et al., 2011; Davis and Caldeira, 2010; Lenzen et al., 2012): decomposing consumption-based pressures into scale and impact intensity components separately for domestic and foreign production identifies a dual role of imports that has received limited attention in sustainability assessments based on planetary boundaries. A rising share of imported goods shifts the consumption bundle toward production technologies that are, on average, more impact-intensive than domestic ones, slowing aggregate efficiency gains even where intensities fall within both origins, while import growth offsets aggregate efficiency gains altogether. Third, the paper speaks to the literature decomposing changes in pollution into scale, composition, and technique effects (e.g., Grossman and Krueger, 1993; Copeland et al., 2022; Levinson, 2009; Shapiro and Walker, 2018). We add by bringing this decomposition to a consumption-based approach. Unlike these studies, we combine composition and technique in a single impact intensity and reserve the term composition for shifts between domestic and foreign production. We highlight a channel that production-based decompositions cannot detect: the shift of consumption toward more impact-intensive imports. Fourth, we contribute to the contested literature on absolute decoupling (Haberl et al., 2020; Parrique et al., 2019; Vadén et al., 2020; Wiedmann et al., 2020; Hubacek et al., 2021) by showing that, when focusing on the domestically produced part of consumption, absolute decoupling appears feasible—an impression that vanishes once imports are accounted for.

From a policy perspective, our findings highlight the limits of sustainability strategies that focus narrowly on improving domestic production efficiency. We show that, as consumption increasingly shifts toward imported goods with higher associated environmental impact intensity, efficiency gains achieved within domestic production fail to translate into improved

sustainability outcomes. Addressing this disconnect requires complementing territorial accounting with supply-chain measures and policy instruments that explicitly account for the scale and composition of consumption. The German case illustrates this challenge across a broad range of environmental impact categories, and our international comparison shows that it is shared by open, high-income economies more generally.

2 Results

2.1 Sustainability of the German economy

We measure Germany's consumption-based impacts—all impacts along the global supply chains of goods and services consumed in Germany, wherever they occur—including direct use-phase impacts, which arise when goods are used, for example from heating and private vehicles. We compare them with Germany's equal-per-capita share of the planetary boundaries, translated into life-cycle impact indicators by Sala et al. (2020) (Section 4). We find, first, that boundary transgressions are widespread and substantial. Germany exceeds national sustainability thresholds in six of the seven impact categories over 1995–2022; only water use remains within the boundary (Fig. 1). The magnitude of transgression differs sharply across the remaining categories: particulate matter, climate change, and fossil resource use exhibit severe overshooting throughout the period, whereas mineral resource use, land use, and photochemical ozone formation are closer to the threshold and overshoot moderately.

Second, transgressions are persistent. Sustainability performance shows little improvement over time, with most series remaining largely flat (Fig. 1). Where impacts declined between 1995 and 2022, the decline was modest, reaching at most 28% (Table 1, Column (1)). In some categories, we even find sizable increases: mineral resource use and water use increase by more than 130%. Both impact categories, however, start from a low initial level and therefore remain close to or below the boundary threshold. In sum, these figures show that, over nearly three decades, German consumption has not converged toward absolute sustainability in most impact categories.

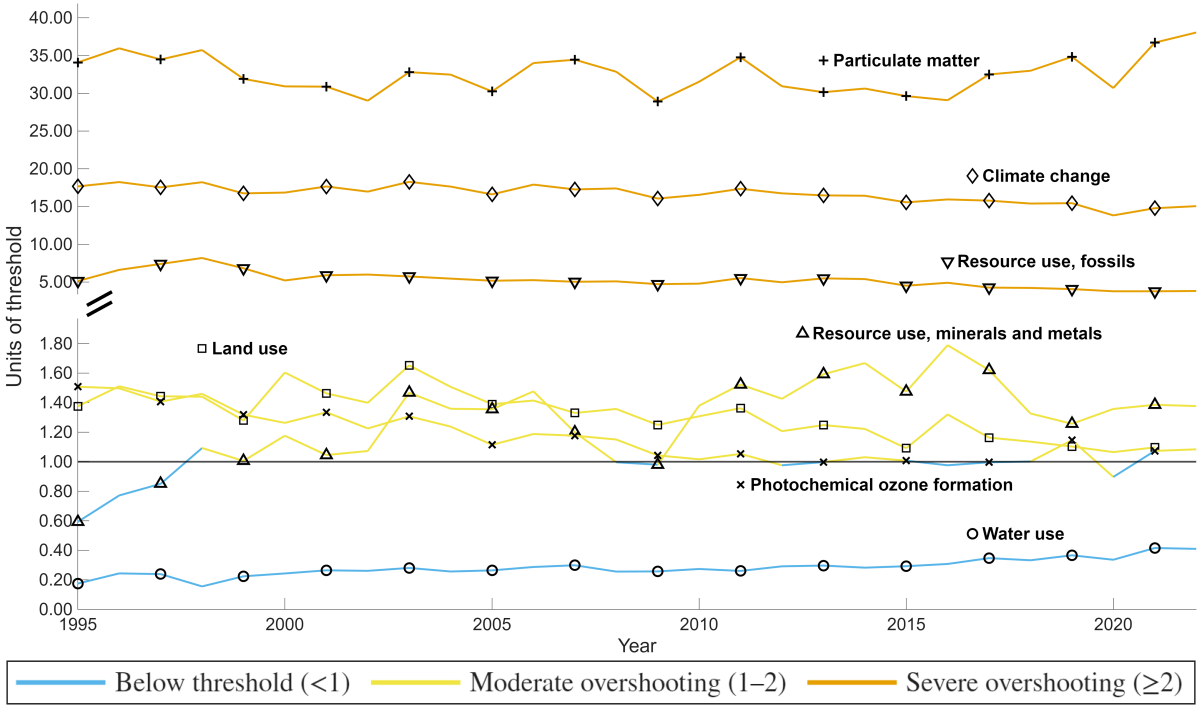


Fig. 1 | Germany's consumption-based impacts relative to planetary-boundary thresholds. Each line shows Germany's consumption-based impacts in one category, including use-phase impacts, over 1995–2022, normalized to the category's threshold so that the horizontal line at 1.00 marks the boundary. Color follows a traffic-light scheme: blue where the threshold is met (below 1.00), yellow where it is moderately exceeded (1.00–2.00), and orange where it is severely exceeded (above 2.00). The vertical axis is compressed above 2.2 to show the categories that exceed their threshold several-fold. Section 4.4 describes the thresholds.

Table 1 | Long-run growth rates by impact category, 1995–2022.

	Total (%) (1)	Domestic production (%) (2)	Foreign production (%) (3)
Climate change	−14.8	−44.2	15.1
Particulate matter	11.6	−30.5	49.0
Photochemical ozone formation	−28.1	−54.5	1.2
Land use	−20.3	−41.7	−19.2
Water use	134.2	−55.0	135.5
Resource use, fossils	−25.7	−84.2	−5.4
Resource use, minerals and metals	132.2	31.0	132.8

Cumulative growth rates. Column (1): Germany's consumption-based impacts, including use-phase impacts (the series in Fig. 1). Columns (2) and (3): their domestic and foreign components (Fig. 2).

2.2 Geographic decomposition

To better understand why aggregate sustainability performance changed so little, we decompose impacts into their domestic and foreign components. We split the impacts of producing for German final demand by where they occur: in Germany or abroad.

In every impact category, impacts generated abroad exceed those generated in Germany in every year, with the sole exception of climate change in 1995 (Fig. 2). Moreover, the environmental pressures attributable to German consumption are increasingly generated abroad: domestic-production impacts decline in almost all categories (Table 1, Column (2)). The exception is mineral resource use, which rose by 31%. By contrast, foreign-production impacts increase in almost all categories (Table 1, Column (3)). Only land use and fossil resource use display a limited long-run decline, and photochemical ozone formation is roughly flat.

The geographic decomposition clarifies why aggregate sustainability indicators remain largely flat: improvements in domestic production are counterbalanced by the larger and increasingly dominant impacts embodied in imports.

2.3 Decomposition into scale and impact intensity

To distinguish whether environmental pressures arise primarily from the growing scale of production for German final demand or from the impact intensity of that production—the impact embodied per euro of production—we separate impacts by origin $O \in \{D, F\}$, where F and D indicate foreign and domestic production. For each origin, total impacts can be written as

$$\begin{aligned} Impact_O &= \sum_i e_{iO} = \sum_i f_{iO} v_{iO} \\ &= \underbrace{V_O}_{\text{scale}} \underbrace{\sum_i f_{iO} \frac{v_{iO}}{V_O}}_{\text{impact intensity}}, \end{aligned} \tag{1}$$

where f_{iO} denotes the impact per euro of production in sector i of origin O that is destined for German final demand—the sector-country-specific impact intensity. The variable v_{iO} denotes

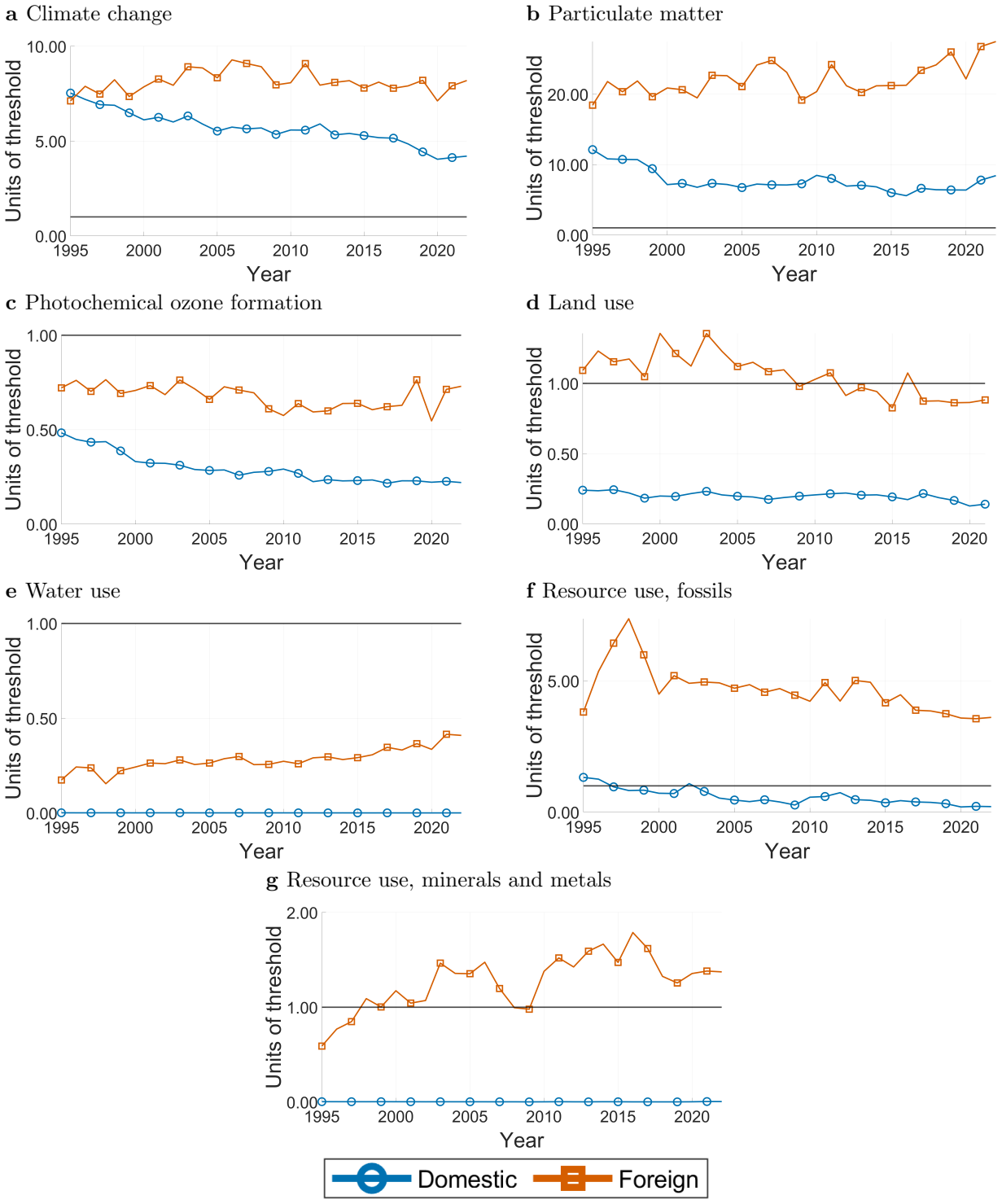


Fig. 2 | Geographic decomposition. Germany's consumption-based impacts in each category, in the impact units of Table 5, split by where each production step takes place: in Germany (domestic) or abroad (foreign). Use-phase impacts are omitted because they are small relative to both components; the two components plus use-phase impacts sum to the series in Fig. 1 before normalization. Section 4.2 describes the use phase.

the corresponding total value of production in sector i in O for German final demand, and V_O is the total value of production from origin O for German final demand. Equation (1) shows that total impact can be separated into a scale and an impact intensity component, where impact intensity is an output-weighted average across sectors (Appendix 7).

Two choices shape our measures. First, in contrast to the literature (Levinson, 2009; Shapiro and Walker, 2018), we keep composition and technique in a single output-weighted impact intensity. A separate technique effect would hold the consumption basket fixed at a base period; it would therefore depend on the base year and become uninformative when consumption shifts away from a sector, since efficiency gains in goods no longer consumed do not affect realized impact. Second, we deflate the scale and impact-intensity components with origin-specific price indices: the German gross domestic product (GDP) deflator for domestic production, the import price index for imports, and, for the combined consumption basket, the average of the two weighted by the nominal shares of domestic and foreign production for German final demand. Without deflation, price changes would bias the trends in both components: inflation would overstate the growth in scale and appear as an improvement in impact intensity, since the same impact would be divided by a higher nominal value of production. As we show in Appendix 7.2, sector-country price movements cancel from the impact-intensity term under the assumption that the sector-country price embedded in the value of production for German final demand equals the price embedded in total sectoral output.

Hence, our measures of impact intensity reflect efficiency improvements over time net of price changes, in line with the literature that isolates a technique effect (Levinson, 2009; Shapiro and Walker, 2018). The comparison of foreign and domestic production, however, is not adjusted for price-level differences between origins, which is why we refer to impact per euro of production for final demand as impact intensity rather than technology. To the extent that imports originate in lower-price economies, price-level differences inflate foreign relative to domestic intensity. The cross-origin gap we report should therefore be read as partly reflecting prices rather than physical technique alone.

In all impact categories, impact intensity is substantially higher for foreign than for domestic production (Fig. 3b–3h). At the same time, the scale of domestic production for

German final demand is larger than that of foreign production, although the gap narrows and the import share rises (Fig. 3a). Both orderings are unchanged at current prices (for intensities, see Table 10). The dominance of foreign-production impacts (Fig. 2) therefore reflects systematically higher impact intensities rather than larger import volumes.

This intensity gap together with the rising import share shapes how aggregate impacts evolve over time, which we examine by tracing the growth of scale and impact intensity by origin since 1995 (Fig. 4). Over this period, the real scale of production for German final demand has grown, with imports rising substantially faster than domestic production (Fig. 4a). At the same time, impact intensity has fallen both abroad and at home (Fig. 4b–4h), apart from a spike at the end of the period in mineral resource use and water use. Foreign intensity in mineral resource use is the only series without a decline over the full period: it rose until the mid-2010s and has since fallen back to about its 1995 level. For imports, these efficiency gains are offset by the growth in scale, so that import-related impacts rise in most categories (Fig. 2). For domestic production, efficiency gains outpace the growth in scale, so that domestic impacts fall in absolute terms in all categories but mineral resource use, which is evidence of absolute decoupling when focusing on domestic production alone.

Overall, efficiency improvements occur at broadly similar rates in domestic and foreign production. However, the impact intensity of the combined consumption basket (*Aggregate* in Fig. 4) falls more slowly, or rises, relative to impact intensity within either origin in all categories, with one temporary exception for land use in 2010–2020. This result is explained by the level differences in impact intensities and the rising share of imported goods in the German final demand basket. As a result, even parallel improvements in domestic and foreign impact intensity will not translate into aggregate sustainability gains as long as consumption shifts toward production systems that remain more impact-intensive on average.

Taken together, the persistence of boundary transgressions does not stem from a lack of efficiency gains: impact intensities fall in both origins in all categories, with the one exception noted above. Rather, these gains are outpaced by the growing scale of consumption and its shift toward more impact-intensive imports.

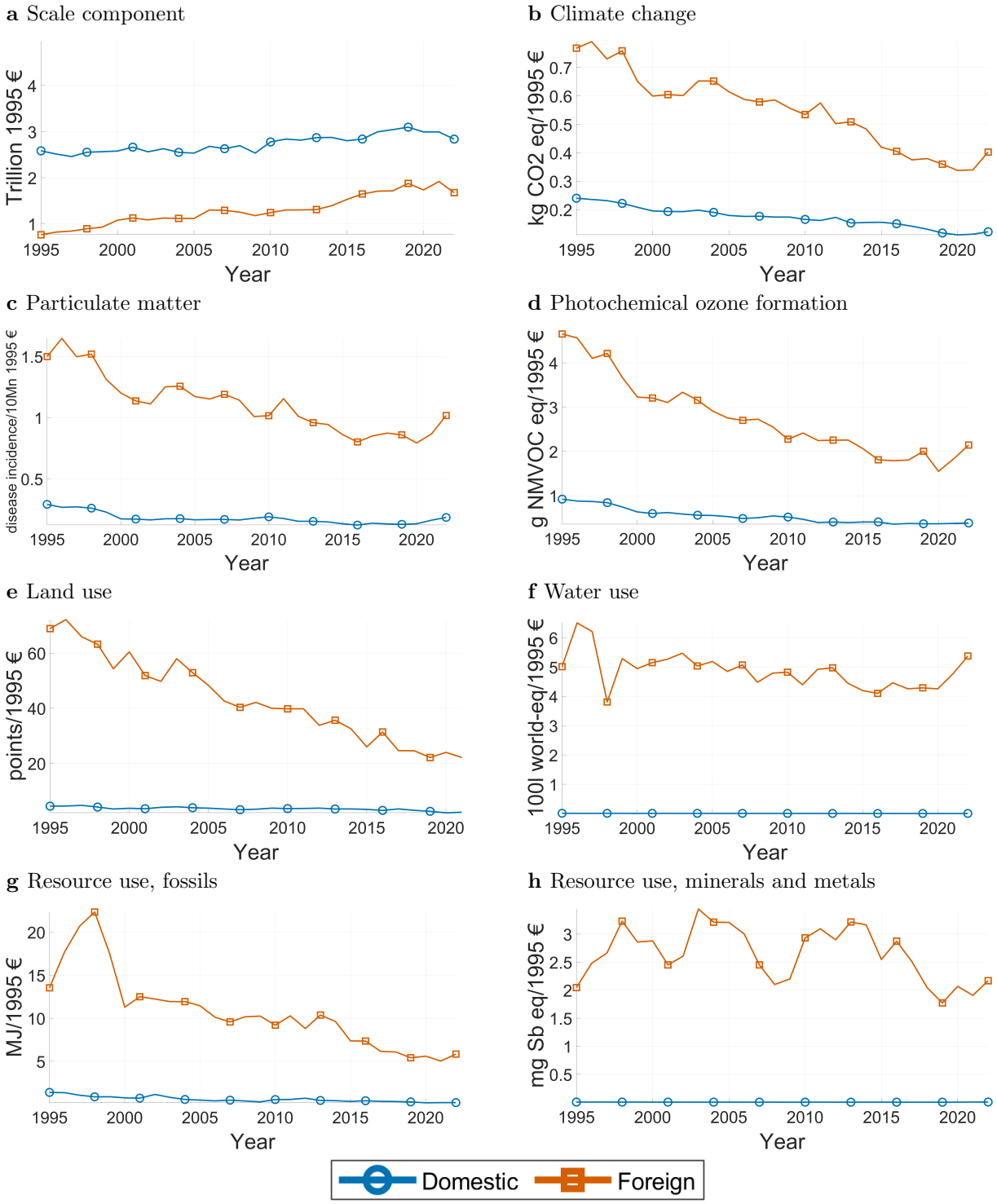


Fig. 3 | Scale and impact intensity of domestic and foreign production. a, Scale component: the value of production for German final demand by origin, in trillion euros at 1995 prices; it is common to all categories. **b**–**h**, Impact intensity of domestic and foreign production in each category, in impact units per euro of production at 1995 prices. Domestic production is deflated with the German GDP deflator and foreign production with the import price index (Appendix 7.2).

2.4 Contribution of the rising import share

To quantify the contribution of the shift toward imports on its own, we ask how the impact trajectories would have differed had the share of imports in the value of production for German final demand not risen. We construct a constant-import-share counterfactual—an accounting exercise that holds this share at its 1995 level while letting the total value of production and origin-specific intensities evolve as observed (Section 4.6). The experiment can be read as asking how environmental impact would have evolved if, of each euro of production for German final demand, the amount needed to hold the import share at its 1995 level had been spent on the domestic bundle rather than on imports, all else equal.

We find that the rising import share raised cumulative 1995–2022 impact growth by 12.9 (climate change), 16.2 (photochemical ozone formation), 22.9 (land use), 26.9 (particulate matter), 28.0 (fossil resource use), 95.7 (mineral resource use), and 96.8 (water use) percentage points relative to the constant-import-share counterfactual. The smallest increase is observed for the climate change category. The largest effects arise for water and mineral resource use, where domestic production is nearly impact-free, so the rising import share contributes most to cumulative impact growth. The effect operates through the foreign–domestic gap in impact intensity: because each euro reallocated from domestic to foreign production carries more impact, a rising import share raises aggregate impact even as intensities fall within both origins (Appendix 7.4).

2.5 Beyond Germany: international comparison

Is this mechanism a German idiosyncrasy or a general feature of open, high-income economies? To answer this question, we repeat the analysis for the other economies covered individually in EXIOBASE, using each country's own final demand and its equal-per-capita share of each boundary (Appendix 9).

Three findings carry over to nine comparable open, high-income economies (Table 2). First, transgression is the norm: each peer consistently exceeds four to six of the seven boundaries (in at least 80% of years) and lies above threshold in 71–86% of all category-year observations; Germany, at six boundaries and 81%, sits in the middle of this group. Second, foreign production carries more impact per euro of production value than domestic production in almost every category-year: in 100% of them for Germany and six of the nine peers, in 99% for the Netherlands and the United States, and in 88% for Spain, with virtually identical shares at constant prices (Appendix 9). Third, the rising import share raised cumulative 1995–2022 impact growth in every category for every one of these ten economies (70 of 70 country-category cells). Germany is thus representative in the sign of every component of the mechanism. In magnitude, however, it lies at the upper end: across the ten economies, the effect ranges from 3.5 to 96.8 percentage points, the maximum being Germany's effect for water use. Germany's unweighted average effect across categories, 42.8 percentage points, exceeds that of every peer (16–35 percentage points; Table 2). Across all 43 economies, the wealthiest exceed the most boundaries, and in the most years. Imports carry more impact per euro of production value than domestic production in more than half of all category-years in 40 of them; the exceptions are a small group of middle-income or resource-rich economies (Appendix 9.2).

3 Discussion

German consumption has exceeded six of seven planetary-boundary thresholds since the mid-1990s, with little progress toward absolute sustainability. Our results suggest that this persistence reflects a mechanism operating across all seven categories. Impact intensities have fallen at largely similar rates at home and abroad, but these gains have been offset

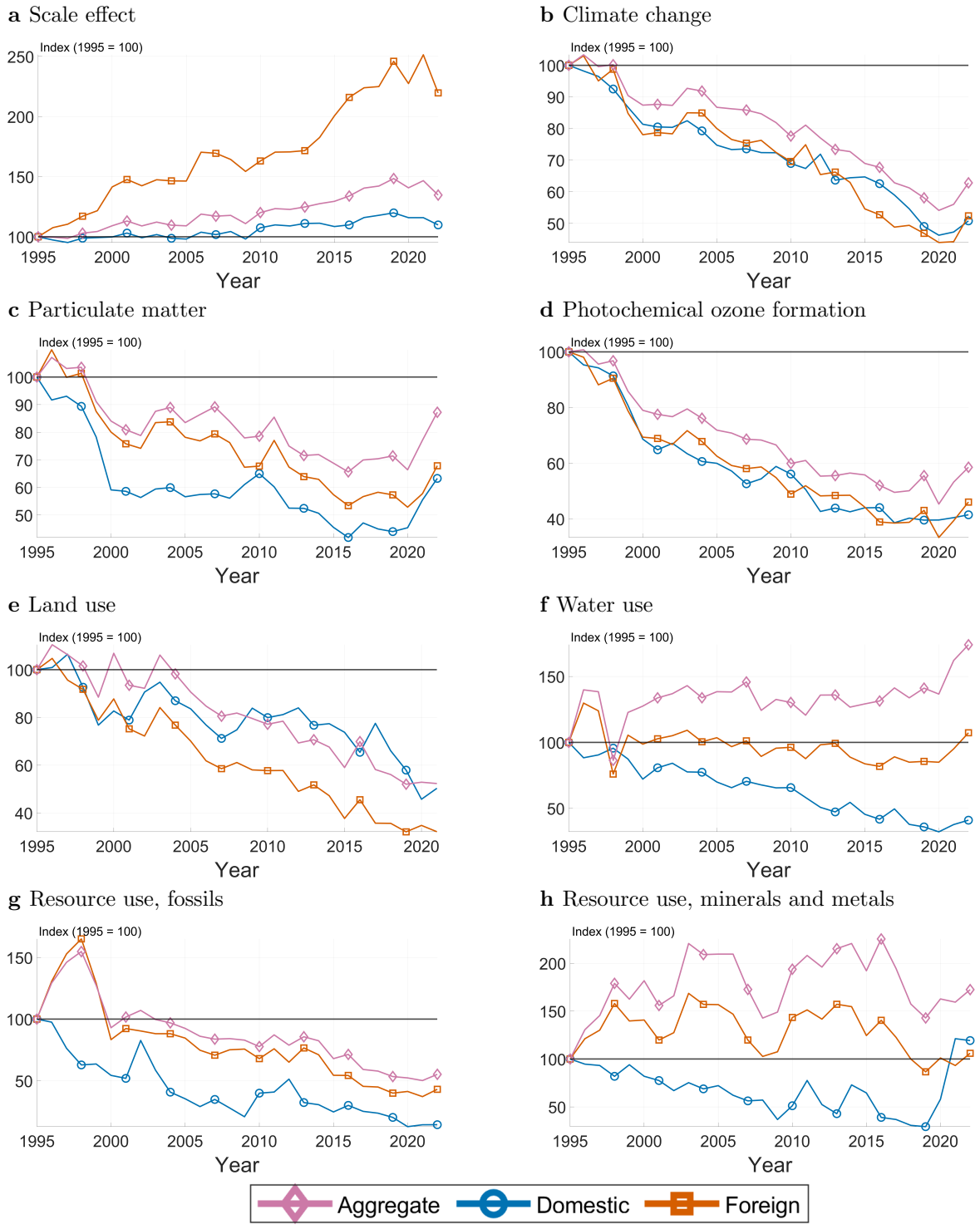


Fig. 4 | Growth of scale and impact intensity since 1995. All series are indexed to 100 in 1995. **a**, Scale effect: how impacts would have evolved had impact intensities stayed at their 1995 levels while the value of production for German final demand, at 1995 prices, followed its observed path; it is common to all categories. **b**–**h**, Impact-intensity effect in each category: how impacts would have evolved had the scale component stayed at its 1995 level. Each panel shows domestic production, foreign production, and the combined basket (aggregate). Domestic production is deflated with the German GDP deflator, foreign production with the import price index, and the combined basket with their weighted average (Appendix 7.2); Appendix 7 derives the decomposition.

Table 2 | The mechanism beyond Germany: comparable open, high-income economies.

Country	Boundaries consistently exceeded (of 7) (1)	Category-years above threshold (%) (2)	Category-years with foreign intensity > domestic (%) (3)	Effect of rising import share on impact growth (pp, min–max) (4)
BE	6	85.7	100	16.6–55.6
US	6	85.7	99	7.3–32.4
NL	6	84.2	99	14.4–78.2
GB	5	81.1	100	5.1–46.3
DE	**6**	**80.6**	**100**	**12.9–96.8**
KR	5	78.6	100	9.0–48.5
JP	5	76.5	100	10.2–66.7
FR	4	75.0	100	6.4–52.0
ES	4	74.0	88	3.5–91.1
IT	4	70.9	100	6.4–39.3

Germany (bold) and nine comparable open, high-income economies, each analyzed with its own final demand and its equal-per-capita share of each planetary boundary. Column (1): number of the seven impact categories in which the country's consumption-based impacts exceed its national threshold in at least 80% of the years 1995–2022. Column (2): share of all category-year observations ($7 \times 28 = 196$) in which the threshold is exceeded. Column (3): share of category-years in which foreign production for the country's final demand carries more impact per current euro than domestic production. Column (4): smallest and largest value, across the seven categories, of the increase in cumulative 1995–2022 impact growth attributable to the rising import share, in percentage points (pp), relative to a counterfactual holding the 1995 import share fixed (Section 2.4); the effect is positive in every category for every country shown. Countries are sorted by Column (2); country codes follow ISO 3166-1 alpha-2. Full country tables for all 43 economies are in Appendix 9.

by the growing scale of consumption and a shift toward imports, which carry more impact per euro than domestic production. A counterfactual holding the import share at its 1995 level shows that this shift accounts for a substantial part of the lack of improvement across categories. The mechanism is not specific to Germany but a structural feature of economies whose consumption relies heavily on international production networks. Comparable open, high-income economies show the same pattern: foreign production is more impact-intensive than domestic production in the large majority of category-years, and the rising import share raised impact growth in every category (Table 2).

Policies that target domestic production efficiency rather than absolute pressures therefore fall short on two counts: efficiency gains are too slow to offset rising consumption, and a growing share of pressures lies abroad, beyond the reach of domestic policy. Our results thus underscore the need for strategies that move beyond efficiency alone and address the

scale and composition of consumption—including supply-chain instruments such as carbon border adjustments and mandatory due diligence—to align economic activity with absolute environmental limits.

Several limitations should be borne in mind. First, our detailed decomposition focuses on Germany. The international comparison shows that the sign of every component of the mechanism is the same across comparable open, high-income economies, but the magnitude of boundary transgressions and of the import-share effect differs across countries. We document this heterogeneity without explaining it. Plausible drivers include country-specific consumption patterns, resource endowments, and trade structures. Second, our decomposition relies on input–output data capturing average production technologies at the country–sector level, and does not allow us to distinguish technologies used for domestic consumption versus exports within a sector. Third, we measure impact intensity per euro of production. Our deflation removes price changes over time within each origin, so that changes over time reflect efficiency rather than prices; any comparison of intensity levels—in particular between foreign and domestic production—also reflects price-level differences. Fourth, the counterfactual is an accounting exercise: it reallocates production values at observed intensities and holds prices, technologies, and the structure of domestic production fixed. Future work with richer data and models could address these caveats—for example, by explaining the cross-country differences, by using higher-resolution production and price data, or by simulating the counterfactual in a general-equilibrium model—and thereby help assess the robustness and policy relevance of these findings more fully.

4 Methods

Our analysis combines three components: environmentally extended multi-regional input–output (EE-MRIO) tables, which trace environmental pressures along global value chains; characterization factors (CFs) from life-cycle impact assessment (LCIA) methods, which translate heterogeneous stressors into comparable impact indicators; and absolute thresholds derived from the planetary-boundaries framework, against which we evaluate these impacts. Technical details are provided in Appendix 6, and Appendix 5 lists all abbreviations.

4.1 Environmentally extended input–output data

We use EXIOBASE 3.9.5 (Stadler et al., 2018), a widely used EE-MRIO database covering 44 countries, five rest-of-world (RoW) regions, and 163 sectors per country for the period 1995–2022. EXIOBASE combines a monetary input–output (IO) system—recording inter-industry flows, final demand, and value added—with environmental satellite accounts that document physical flows associated with production. Our analysis draws on four EXIOBASE environmental extensions relevant for LCIA-based impact assessment: (i) air emissions, (ii) water use, (iii) material use, and (iv) land use. Together, these extensions contain 705 distinct stressors.

To integrate environmental stressors into the IO framework, we express them as intensities, measured as stressor amounts per euro of sectoral output. Combining stressor intensities with the Leontief inverse provides us with the total environmental pressures embodied in consumption, including all upstream supply-chain stages.

4.2 Consumption-based accounting and use-phase impacts

There are two complementary sets of environmental impact accounts commonly used in environmental accounting. Production-based impact accounts attribute environmental pressures to the location where production occurs, irrespective of where goods are ultimately consumed. Consumption-based impact accounts, by contrast, attribute environmental pressures to the location of final consumption. Final consumption here comprises all final demand except exports, that is, consumption by households and the government as well as investment. Pressures embodied in exports are therefore excluded, while those embodied in imports are included. Importantly, the consumption-based account captures not only the pressures from the final production step, but also all upstream pressures generated along international value chains.

We focus on consumption-based sustainability because it aligns with welfare-based interpretations of environmental responsibility. Furthermore, it avoids misleading improvements that arise when production of environmentally intensive consumption goods is relocated abroad rather than reduced. In open economies with deep global value-chain integration,

consumption-based accounts therefore provide a more informative benchmark for assessing sustainability outcomes related to welfare.

We incorporate use-phase impacts, which are generated during the use of goods and services, such as emissions from residential heating, electricity use, or private mobility. These impacts arise directly from final consumption and are reported in a separate extension for the final demand matrix in EXIOBASE. The consumption-based accounting literature often treats use-phase impacts as part of the consumption-based account itself (Wood et al., 2018; Stadler, 2021). We instead treat use-phase impacts separately from the consumption-based account. In the terminology of that literature, our consumption-based account plus use-phase impacts (the aggregate series in Fig. 1) corresponds to the consumption footprint, comprising direct (use-phase) and indirect (consumption-based) impacts. Use-phase impacts are small relative to production-related impacts and decline over time in most categories: their contribution to total impact averages 6.70% across years and categories and peaks at 20.84% for photochemical ozone formation in 2001. We therefore do not analyze them further in the decompositions.

4.3 Impact categories and characterization factors

Planetary boundaries define biophysical limits for critical Earth-system processes (Rockström et al., 2009; Steffen et al., 2015). However, these limits are expressed in physical control variables that are not directly compatible with IO-based accounting. We therefore rely on LCIA-based planetary boundaries developed by Sala et al. (2020), which translate boundary control variables into LCIA indicators and provide corresponding threshold values for environmental impacts. Due to their compatibility with IO frameworks, LCIA impact categories allow for a systematic attribution and tracking of environmental pressures associated with economic activity across sectors, regions, and impact dimensions.

CFs convert heterogeneous stressors into standardized impact units, allowing environmental pressures to be compared and related to impact categories. For example, greenhouse gases are expressed in CO_2-equivalents, particulate matter precursors in expected disease incidence, and land occupation in soil-quality points based on the LANCA (LANd use indicator value CAlculation) model. We use CFs from the Environmental Footprint (EF) 3.1 method provided

by Andreasi Bassi et al. (2023). Characterized impacts are obtained as:

$$\mathbf{P}_t = \mathbf{CF} \times \mathbf{Q}_t,$$

where $\mathbf{Q}_t$ denotes the stressor matrix and $\mathbf{CF}$ the matrix of CFs. For water and land use, CFs vary by country of production because environmental damages depend on local conditions; we therefore construct region-specific CFs for the RoW regions in EXIOBASE.

We retain the seven impact categories for which LCIA-based planetary-boundary thresholds exist and EXIOBASE stressors can be consistently characterized using EF 3.1. The categories are: climate change, particulate matter, photochemical ozone formation, land use, water use, resource use (fossils), and resource use (minerals and metals). Threshold values and units are reported in Table 5 in Appendix 6.4. Not all stressors in our data can be matched to an impact category, since some lack characterization factors and are necessarily left uncharacterized. Appendix 8 shows that including them would not overturn our main results.

4.4 Planetary-boundary thresholds

We assess sustainability by comparing environmental impacts to national thresholds, which are obtained by downscaling global LCIA-based planetary boundaries. We start by collecting global thresholds taken from Sala et al. (2020) and Sanyé-Mengual and Sala (2023); for particulate matter, we recompute the threshold with the average global population over 1995–2022 instead of the 2010 value, which lowers it slightly from 5.16×10^5 to 5.13×10^5 disease incidence (Appendix 6.4.1). We scale global thresholds to a national level using an equal-per-capita approach:

$$\mathrm{PB}_c = \bar{\omega}_c \cdot \mathrm{PB}_{\mathrm{global}},$$

where PB_c and $\mathrm{PB}_{\mathrm{global}}$ are the national and global thresholds and $\bar{\omega}_c$ is the country's average population share over 1995–2022 (United Nations, 2024). Using average shares prevents changes in population from mechanically affecting the sustainability indicator over time. We adopt an equal-per-capita allocation as a transparent and conservative benchmark. Alternative allocation principles based on historical responsibility or mitigation capacity would imply even stricter thresholds for high-income countries such as Germany. Appendix 6.4.2 discusses

these alternatives.

4.5 Trade decomposition

To separate domestic and international sources of environmental impacts, we split consumption-based impacts by where each production step takes place. Domestic impacts arise in production in Germany for German final demand, including domestic intermediate inputs into imported goods; foreign impacts arise abroad at any stage of the supply chain serving German final demand, including foreign intermediate inputs into goods finished in Germany (Appendix 6.1.5). This decomposition allows us to distinguish whether sustainability gaps are driven primarily by domestic or foreign production.

4.6 Scale–intensity decomposition and counterfactual

For each origin $O \in \{D, F\}$, we write impacts as the product of the value of production for German final demand, V_O (*scale*), and the output-weighted average impact per euro of that production (*impact intensity*). V_O is a production-side magnitude and excludes trade and transport margins and net taxes on products. We keep composition and technique in a single impact-intensity measure, because a separate technique effect—the efficiency change that would occur if the consumption basket remained fixed at a base period—depends on the choice of base year and becomes uninformative when consumption shifts away from a sector. We deflate domestic production with the German GDP deflator and foreign production with the import price index (Appendix 7.2), so that changes over time are not driven by price changes; cross-origin differences in intensity levels also reflect price-level differences.

The constant-import-share counterfactual holds the nominal share of foreign production in the value of production for German final demand at its 1995 level, while letting the total value of production and within-origin intensities evolve as observed (Appendix 7.4). The experiment is designed as a partial-equilibrium reallocation rather than a structural simulation: displaced imports are met by domestic production at observed intensities, with no adjustment of prices or technologies. Unlike the scale–intensity decomposition, which tracks changes over time and is therefore deflated, the counterfactual is constructed from nominal production values,

shares, and intensities. This reproduces observed physical impacts exactly without relying on aggregate deflators or the price assumption of Appendix 7.2. Use-phase impacts are included as observed. Fig. 5 in Appendix 7.4 compares observed and counterfactual impacts for all categories.

4.7 International comparison

To assess whether the German results generalize, we replicate the analysis for every country covered individually in EXIOBASE. The replication alters only two country-specific inputs while leaving the accounting framework, the characterization of stressors, and the region-specific characterization factors unchanged: the final demand block, so that domestic and foreign production are defined relative to the country in question; and the national threshold, computed with the same equal-per-capita allocation as for Germany using the country's average population share over 1995–2022. We exclude Taiwan, for which World Bank indicators are not available; all measures are therefore reported for 43 economies. Table 2 reports nine comparable open, high-income economies alongside Germany. Appendix 9 reports the full country tables.

We also check the comparison of foreign and domestic intensity in constant prices, using each country's GDP deflator from the World Bank and its deflator of imports of goods and services from the European Commission's AMECO database; the availability of import price data limits this check to 36 economies (Appendix 9.2).

Data availability

This study uses publicly available data. The environmentally extended multi-regional input–output database EXIOBASE 3.9.5 is available from Zenodo (Stadler et al., 2025). Characterization factors of the Environmental Footprint 3.1 method (Andreasi Bassi et al., 2023) are available from the European Platform on Life Cycle Assessment of the European Commission's Joint Research Centre. Global planetary-boundary thresholds are taken from Sala et al. (2020) and Sanyé-Mengual and Sala (2023). Population data are from the United Nations World Population Prospects 2024 (United Nations, 2024); GDP at purchasing power parity, used

to weight characterization factors for the rest-of-world regions, is from International Monetary Fund (2025). The German import price index is from Statistisches Bundesamt (2025), GDP deflators for Germany and the other countries are taken from World Bank (2025a), and country-specific deflators of imports of goods and services (EUR/ECU basis) for the international comparison are from the European Commission's AMECO database (European Commission, Directorate-General for Economic and Financial Affairs, 2026). The ordering by trade openness in Tables 10 and 11 is based on World Bank (2025b). The processed data underlying this article will be made publicly available upon publication and are available to editors and reviewers on request.

Code availability

All analyses were conducted in MATLAB (R2025b, version 25.2). All code used in this study, which reproduces all figures and tables, will be made publicly available upon publication and is available to editors and reviewers on request.

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Author contributions

S.D. was lead author. S.D., C.K., A.K., L.S., K.T., conceptualized and designed the research; G.M. and K.T. performed the research and analyzed the data; S.D. and K.T. wrote the initial paper draft; S.D., C.K., G.M., L.S., and K.T. reviewed and edited the paper.

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The authors declare no competing interests.

5 Abbreviations

Table 3 | Abbreviations.

Abbreviation	Definition
ADP	abiotic depletion potential
AMECO	annual macro-economic database of the European Commission
CBA	consumption-based account
CF	characterization factor
EE-MRIO	environmentally extended multi-regional input–output
EF 3.1	Environmental Footprint 3.1
GDP	gross domestic product
GRAS	generalized RAS
GWP100	global warming potential over a 100-year horizon
IO	input–output
LANCA	LANd use indicator value CAlculation
LCA	life-cycle assessment
LCIA	life-cycle impact assessment
NMVOC	non-methane volatile organic compounds
pp	percentage points
PPP	purchasing power parity
RAS	biproportional matrix-balancing method
RoW	rest of the world

Country codes follow ISO 3166-1 alpha-2 (for example, DE, Germany; GB, United Kingdom).

6 Supplementary methods

6.1 Environmentally extended input–output tables

We use the EE-MRIO system of EXIOBASE 3.9.5 (Section 4.1). Its five RoW regions are Asia and Pacific, America, Europe, Africa, and the Middle East.

6.1.1 Input–output tables

IO tables describe the flows of intermediate and final goods across sectors and regions. Let there be m countries (or regions) and k sectors per country, implying $n = mk$ country–sector combinations. The core monetary data consist of:

- a transaction matrix $\mathbf{Z} \in \mathbb{R}^{n \times n}$, where z_{ij} denotes intermediate sales from sector i to sector j;
- a final demand matrix $\mathbf{Y} \in \mathbb{R}^{n \times m}$, where y_{ij} denotes final demand in country j for the output of sector i (private and public consumption and investment);

- a value-added vector $\mathbf{VA} \in \mathbb{R}^{1\times n}$, where va_j denotes value added in sector j;
- a total output vector $\mathbf{x} \in \mathbb{R}^{n\times 1}$, where x_i denotes total output of sector i.

Formally,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{Z} = \begin{bmatrix} z_{11} & \dots & z_{1n} \\ \vdots & \ddots & \vdots \\ z_{n1} & \dots & z_{nn} \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} y_{11} & \dots & y_{1m} \\ \vdots & \ddots & \vdots \\ y_{n1} & \dots & y_{nm} \end{bmatrix}.$$

For sector i, total output equals the sum of intermediate and final sales:

$$x_i = \sum_{j=1}^{n} z_{ij} + \sum_{j=1}^{m} y_{ij}.$$

In matrix notation,

$$\mathbf{x} = \mathbf{Z}\mathbf{i}_n + \mathbf{Y}\mathbf{i}_m,$$

where $\mathbf{i}_n$ and $\mathbf{i}_m$ are summation vectors of ones of appropriate dimension. The total input to sector j equals the sum of all intermediate inputs plus value added:

$$x_j^{\text{input}} = \sum_{i=1}^{n} z_{ij} + va_j.$$

By construction, IO tables satisfy $x_j^{\text{input}} = x_j$ for all sectors j.

The technical coefficient matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ has elements

$$a_{ij} = \frac{z_{ij}}{x_j},$$

and captures the direct input requirements per unit of output of sector j. In matrix notation,

$$\mathbf{A} = \mathbf{Z}\widehat{\mathbf{x}}^{-1},$$

where $\widehat{\mathbf{x}}$ denotes the diagonal matrix with $\mathbf{x}$ on its diagonal. Total requirements, including all direct and indirect inputs, are captured by the Leontief inverse

$$\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}.$$

Given a vector of final demand $\mathbf{y}$ (e.g., $\mathbf{y} = \mathbf{Y}\mathbf{i}_m$), total output is

$$\mathbf{x} = \mathbf{L}\,\mathbf{y}.$$

Block-diagonalization of final demand For the accounting formulas below, it is convenient to represent final demand in block-diagonal form. We define an operator $\text{diag}^*(\mathbf{Y})$ that builds a block-structured matrix $\mathbf{B}$ from the columns of $\mathbf{Y}$. Formally,

$$\mathbf{B} = \text{diag}^*(\mathbf{Y}) = \begin{bmatrix} \widehat{\mathbf{Y}}_{11} & \dots & \widehat{\mathbf{Y}}_{1m} \\ \vdots & \ddots & \vdots \\ \widehat{\mathbf{Y}}_{m1} & \dots & \widehat{\mathbf{Y}}_{mm} \end{bmatrix},$$

where each block $\widehat{\mathbf{Y}}_{ij}$ is a $k \times k$ diagonal matrix with entries corresponding to the final demand vector of country j for the k sectors of country i.

6.1.2 Environmental extensions

Environmental extensions augment the monetary IO system with satellite accounts. EXIOBASE provides such satellite accounts in the form of a stressor matrix $\mathbf{Q}$:

$$\mathbf{Q} = \begin{bmatrix} q_{11} & \dots & q_{1n} \\ \vdots & \ddots & \vdots \\ q_{s1} & \dots & q_{sn} \end{bmatrix},$$

where q_{sj} denotes the amount of stressor s (e.g., CO_2, land use, water use) associated with the output of sector j. In EXIOBASE, these stressors are initially uncharacterized, that is, not yet translated into units of the LCIA impact categories.

We convert stressors into intensities per euro of output:

$$\mathbf{F} = \mathbf{Q}\,\widehat{\mathbf{x}}^{-1} = \begin{bmatrix} \frac{q_{11}}{x_1} & \dots & \frac{q_{1n}}{x_n} \\ \vdots & \ddots & \vdots \\ \frac{q_{s1}}{x_1} & \dots & \frac{q_{sn}}{x_n} \end{bmatrix},$$

where $f_{sj} = \frac{q_{sj}}{x_j}$ denotes the amount of stressor s per euro of output of sector j. Inserting $\mathbf{x} = \mathbf{L}\mathbf{y}$ attributes stressors to final demand: for stressor s, the s-th row of $\mathbf{FL}\widehat{\mathbf{y}}$ gives the amount embodied in final demand for the output of each sector. The matrix $\mathbf{FL}$ contains the stressor multipliers, that is, the stressors embodied in one euro of final demand for each sector.

6.1.3 Aggregation

Aggregation follows Stadler (2021). Let $\mathbf{N} \in \mathbb{R}^{s \times k}$ be any sectoral matrix (e.g., stressor intensities or impacts) with k sectors. To aggregate sectors into $q < k$ groups, we define a

sectoral aggregation matrix $\mathbf{G}_k \in \mathbb{R}^{q \times k}$:

$$\mathbf{G}_k = \begin{bmatrix} g_{11} & \cdots & g_{1k} \\ \vdots & \ddots & \vdots \\ g_{q1} & \cdots & g_{qk} \end{bmatrix},$$

where each column has exactly one entry equal to 1, which indicates the group to which the original sector belongs, and zeros elsewhere. Aggregated values are obtained as

$$\mathbf{N}_{\text{agg}} = \mathbf{N}\,\mathbf{G}_k^{\top}.$$

Aggregating all $k = 163$ sectors to a single national aggregate corresponds to $\mathbf{G}_k$ being a $1 \times k$ row vector of ones.

6.1.4 Consumption-based accounts and use-phase impacts

Consumption-based account We denote the CBA by $\mathbf{D}_t^{cba}$ for a given year t, computed first for uncharacterized stressors. Using the notation above and the block-diagonalization operator $\text{diag}^*(\cdot)$, it can be written as:

$$\mathbf{D}_t^{cba} = \mathbf{F}_t \mathbf{L}_t \ \text{diag}^*(\mathbf{Y}_t).$$

Each column of $\mathbf{D}_t^{cba}$ contains the stressors embodied in one country's final demand for the products of one sector.

Use-phase impacts Use-phase stressors (Section 4.2) are reported in a separate EXIOBASE extension for final demand. We add them to the CBA at the country level:

$$\mathbf{D}_t^{tot} = \mathbf{D}_t^{cba}\,\mathbf{G}_n^{\top} + \mathbf{F}_{y,t},$$

where $\mathbf{D}_t^{tot}$ is the CBA plus use-phase stressors, $\mathbf{G}_n \in \mathbb{R}^{m \times n}$ sums the columns of $\mathbf{D}_t^{cba}$ by country of final demand, and $\mathbf{F}_{y,t} \in \mathbb{R}^{s \times m}$ contains the use-phase stressors of each country's final demand in year t.

6.1.5 Domestic and foreign components

We split Germany's CBA by the location of each production step. Let $\mathbf{M}^F$ denote an $n \times n$ diagonal trade mask with ones for all foreign country–sector combinations and zeros for

the k German sectors, and let $\mathbf{M}^D = \mathbf{I} - \mathbf{M}^F$. Inserting $\mathbf{I} = \mathbf{M}^D + \mathbf{M}^F$ into the CBA of Appendix 6.1.4 yields

$$\mathbf{D}_t^{cba} = \underbrace{\mathbf{F}_t \, \mathbf{M}^D \, \mathbf{L}_t \, \mathrm{diag}^*(\mathbf{Y}_t)}_{\mathbf{D}_t^D} + \underbrace{\mathbf{F}_t \, \mathbf{M}^F \, \mathbf{L}_t \, \mathrm{diag}^*(\mathbf{Y}_t)}_{\mathbf{D}_t^F},$$

evaluated for the columns of $\mathrm{diag}^*(\mathbf{Y}_t)$ that belong to German final demand. Here, $\mathbf{L}_t \, \mathrm{diag}^*(\mathbf{Y}_t)$ gives the output required in every country and sector to serve this final demand, and the mask $\mathbf{M}^F$ sets the part produced in Germany to zero. The foreign component $\mathbf{D}_t^F$ thus contains all stressors generated abroad, at any stage of the supply chain, and the domestic component $\mathbf{D}_t^D$, all stressors generated in Germany. Use-phase stressors belong to neither component. For characterized impacts, $\mathbf{p}_t$ replaces $\mathbf{F}_t$ (Appendix 6.3.2).

6.1.6 Comparison to process-based life-cycle assessment

Compared with process-based LCA, EE-MRIO analysis covers upstream supply chains more completely. Product-level LCA must draw a finite system boundary and truncates impacts beyond it (Agez et al., 2022), whereas the Leontief inverse includes all upstream stages. EE-MRIO analysis thus reduces truncation error while still attributing impacts to sectors and consumption categories.

6.2 Data cleaning

Before the analysis, we clean the IO data to reduce the influence of extreme observations and to improve their consistency over time, while preserving the accounting identities. Cleaning proceeds in three steps: winsorizing, temporal smoothing, and rebalancing.

6.2.1 Winsorizing the data

We winsorize the technical coefficient matrix $\mathbf{A}$ and the final demand matrix $\mathbf{Y}$. For $\mathbf{A}$, we proceed column by column: for each column j, we compute the 99.5th percentile of the off-diagonal elements $\{a_{ij} : i \neq j\}$ and cap the off-diagonal elements above it at this value; all other elements remain unchanged. We exclude the diagonal elements from the percentile because they are typically much larger and would otherwise dominate it. We verify that all coefficients remain strictly below one.

The final-demand data are cleaned using a share-based approach on a country basis. We create the final-demand shares as $S_{ik} = \frac{Y_{ik}}{\sum_{k'} Y_{ik'}}$. In contrast to levels, shares are scale-invariant and therefore better suited for the winsorizing process. Within each country block,

we winsorize the final-demand shares column by column at the 99.5th percentile, capping unusually large shares as $S_{ik}^{win} = \min(S_{ik}, p_{99.5})$. We then reconstruct the final-demand levels in two steps: we renormalize the winsorized shares to sum to one, $\tilde{S}_{ik} = \frac{S_{ik}^{win}}{\sum_{k'} S_{ik'}^{win}}$, and multiply them by the original row totals, $Y_{ik}^{new} = \tilde{S}_{ik} \sum_{k'} Y_{ik'}$.

6.2.2 Temporal smoothing

Without smoothing, the technical coefficients in EXIOBASE display implausibly large jumps between years. We therefore smooth the winsorized coefficients in two respects. First, we treat an isolated zero, that is, a coefficient $a_{ijt} = 0$ with positive values in $t-1$ and $t+1$, as a data artifact and replace it by linear interpolation between its neighbors. Second, we treat one-period spikes, defined as values exceeding five times the larger of the two adjacent values, as transitory outliers and set them to the average of the adjacent values. We do not smooth longer runs of zeros or trailing zeros.

6.2.3 Rebalancing with generalized RAS

After winsorizing and smoothing, we reconstruct the intermediate transaction matrix from the cleaned technical coefficients and the original gross output vector, $\mathbf{Z}_t^{new} = \mathbf{A}_t^{smooth} \operatorname{diag}(\mathbf{x}_t^{target})$. The resulting system no longer satisfies the IO accounting identities exactly: row and column totals deviate slightly from those of the original system. We restore consistency with the RAS procedure (Miller and Blair, 2009, ch. 9).

The RAS algorithm iteratively alternates between two steps. First, columns are scaled such that their sums match the target column totals. In the second step, rows are scaled such that their sums match the target row totals. These two steps are repeated until the maximum deviation between achieved and target row/column totals falls below the tolerance level. The goal is to re-adjust the cleaned data such that the target totals are matched, while altering individual cells as little as possible in relative terms. Rebalancing is performed on an extended matrix that stacks the transaction matrix and final demand matrix into a single block matrix $\mathbf{R} = \begin{bmatrix} \mathbf{Z} & \mathbf{Y} \end{bmatrix}$. The row and column sums of the pre-cleaning system are treated as fixed targets. The row sums correspond to sectoral gross output, while the column sums represent intermediate input use and final demand totals, respectively.

Since standard RAS rebalancing requires all matrix elements to be non-negative—which is not the case for the final-demand matrix $\mathbf{Y}$—we use generalized RAS (GRAS), developed by Junius and Oosterhaven (2003) and refined by Lenzen et al. (2007) and Temurshoev et al. (2013). GRAS partitions $\mathbf{R}$ into a matrix $\mathbf{R}^+$ of its positive elements and a matrix $\mathbf{R}^-$ holding the absolute values of its negative elements, so that $\mathbf{R} = \mathbf{R}^+ - \mathbf{R}^-$. Rebalancing weights are

applied separately to $\mathbf{R}^+$ and $\mathbf{R}^-$, which are then recombined into the rebalanced $\mathbf{R}$. When all elements of $\mathbf{R}$ are non-negative, GRAS reduces to standard RAS.

6.3 Characterization of environmental stressors

We characterize the stressors of Appendix 6.1.2 in two steps: we match them to CFs (Appendix 6.3.1) and compute impacts in the units of each impact category (Appendix 6.3.2).

Table 4 | EXIOBASE extensions, CF scope, and units by impact category.

Impact category	EXIOBASE extension	CF scope	Unit before characterization	Unit after characterization
	(1)	(2)	(3)	(4)
Climate change	Air emissions	Global	kg	kg CO_2-eq
Particulate matter	Air emissions	Global	kg	disease incidence
Photochemical ozone formation	Air emissions	Global	kg	kg NMVOC-eq
Land use	Land	Country- or region-specific	km^2	Pt (LANCA index)
Water use	Water	Country- or region-specific	Mm^3	m^3 world-eq
Resource use, fossils	Material extraction	Global	kt	MJ
Resource use, minerals and metals	Material extraction	Global	kt	kg Sb-eq

Column (1): EXIOBASE extension used for the category. Column (2): whether EF 3.1 CFs are global or country- or region-specific. Columns (3) and (4): units of stressors before and after characterization. Appendix 6.3.1 describes how CFs are matched to stressors.

6.3.1 Matching characterization factors to EXIOBASE stressors

EF 3.1, the method recommended by the European Commission for Product and Organisation Environmental Footprints (European Commission, 2021; Andreasi Bassi et al., 2023), provides CFs for 16 impact categories, of which we use seven (Section 4.3; Table 4).

EF 3.1 provides CFs for several stressors that are not contained in EXIOBASE and, conversely, EXIOBASE includes stressors that are not characterized in EF 3.1. Following Sala et al. (2020) and their supplementary material, we match CFs and stressors where possible, resulting in coverage comparable to their work. Where no corresponding stressor exists in EXIOBASE, the CF cannot be applied; where a stressor is included in EXIOBASE but has no EF 3.1 CF relevant for our impact categories, we assign a CF of zero. Consequently, our impact estimates should be interpreted as lower bounds of the German economy's true

contribution to each impact category. While incomplete coverage may affect absolute levels in some categories, it is unlikely to alter our qualitative conclusions, as boundary exceedances are large, persistent, and driven by stressors with well-characterized CFs. Appendix 8 assesses the importance of missing characterization factors for our main results.

Characterization acts as an implicit weighting scheme: impacts in a category are unweighted sums of characterized stressors, so the CFs alone determine, for example, the relative weight of CO_2, CH_4, and N_2O in climate change.

For five of the seven categories (climate change, particulate matter, photochemical ozone formation, fossil resource use, and mineral resource use), EF 3.1 CFs do not vary by region. For climate change and resource use, this reflects damage mechanisms that are largely independent of where emissions or extraction occur; for particulate matter and photochemical ozone formation, a single global CF is a modeling simplification.

For land use and water use, EF 3.1 provides country- and region-specific CFs (Section 4.3). To match the five EXIOBASE RoW regions (Appendix 6.1), we average the country-specific CFs within each region, weighting countries by their GDP at PPP. Let CF_j denote the CF of country j in a given impact category and PPP_t^j its GDP at PPP in year t. We first average GDP at PPP over 1995–2022:

$$PPP^j_{\text{avg}} = \frac{1}{T} \sum_{t=1995}^{2022} PPP^j_t,$$

with $T = 2022 - 1995 + 1 = 28$. We then define PPP-based weights within each RoW region:

$$PPP^j_{\text{weight}} = \frac{PPP^j_{\text{avg}}}{\sum_{l=1}^{L} PPP^l_{\text{avg}}},$$

where $l = 1, \ldots, L$ indexes the countries in the region.

$$CF_{\text{RoW}} = \sum_{j=1}^{L} PPP^j_{\text{weight}} \cdot CF^j.$$

We apply this procedure to each RoW region for land use and water use, which yields CFs for all 49 EXIOBASE countries and regions.

6.3.2 Calculation of impacts

Let $\mathbf{Q}_t$ denote the stressor matrix in year t (Appendix 6.1.2) and $\mathbf{CF}_w$ the $s \times n$ matrix of

CFs for impact category w. The impact vector for impact category w is obtained from

$$\mathbf{P}_{t,w} = \mathbf{i}_s^\top (\mathbf{CF}_w \odot \mathbf{Q}_t),$$

where $\mathbf{i}_s$ is an $s \times 1$ summation vector.

For five categories, the CFs are global and collected in the $s \times 1$ vector $\mathbf{cf}_w^{single}$. For land use and water use, they are country- or region-specific (Appendix 6.3.1) and collected in the $s \times m$ matrix $\mathbf{CF}_w^{country}$. Because these CFs do not vary by sector, each of its columns is repeated k times, once for each sector of the country:

$$\mathbf{CF}_w = \begin{cases} \mathbf{cf}_w^{single}\, \mathbf{i}_n^\top & \text{for } w \notin \{\text{land use, water use}\}, \\ \mathbf{CF}_w^{country} \otimes \mathbf{i}_k^\top & \text{for } w \in \{\text{land use, water use}\}. \end{cases}$$

Finally, all impact vectors are stacked to obtain the impact matrix $\mathbf{P_t}$

$$\mathbf{P_t} = \begin{bmatrix} \mathbf{P_{t,1}} \\ \vdots \\ \mathbf{P_{t,W}} \end{bmatrix}.$$

Analogously to the stressor intensities $\mathbf{F}_t$ (Appendix 6.1.2), we define the impact intensity matrix

$$\mathbf{p}_t = \mathbf{P}_t\, \widehat{\mathbf{x}}_t^{-1},$$

where $\widehat{\mathbf{x}}_t$ is the diagonal matrix of total sectoral output in year t. The CBA in impact units follows from replacing $\mathbf{F}_t$ with $\mathbf{p}_t$ in the formula of Appendix 6.1.4:

$$\mathbf{D}_t^{cba} = \mathbf{p}_t \mathbf{L}_t\ \mathrm{diag}^*(\mathbf{Y}_t).$$

6.4 Planetary-boundary thresholds

6.4.1 Global thresholds

We take global thresholds from Sanyé-Mengual and Sala (2023), who compile the LCIA-based planetary boundaries of Sala et al. (2020) (Table 5). The exception is particulate matter, whose threshold depends on global population. For consistency with the population data used for downscaling (Appendix 6.4.2), we recompute it following Sala et al. (2020) with the average global population over 1995–2022 (United Nations, 2024) instead of the 2010 value. The resulting threshold, 5.13×10^5 disease incidence, is slightly below the 5.16×10^5 reported in both sources.

Table 5 | Impact categories, units, and planetary-boundary thresholds.

Impact category	Unit	Underlying model	Global threshold	German threshold
	(1)	(2)	(3)	(4)
Climate change	kg CO_2-eq	Bern model, GWP100	6.81×10^{12}	8.23×10^{10}
Particulate matter	disease incidence	UNEP-recommended disease-incidence model	5.13×10^{5}	6.20×10^{3}
Photochemical ozone formation	kg NMVOC-eq	LOTOS-EUROS	4.07×10^{11}	4.92×10^{9}
Land use	Pt	Soil quality index based on LANCA	3.98×10^{15}	4.81×10^{13}
Water use	m^3 world-eq	AWARE	1.82×10^{14}	2.20×10^{12}
Resource use, fossils	MJ	ADP, fossil fuels	2.24×10^{14}	2.71×10^{12}
Resource use, minerals and metals	kg Sb-eq	ADP, ultimate reserves	2.19×10^{8}	2.65×10^{6}

Global thresholds are from Sanyé-Mengual and Sala (2023), except for particulate matter (Appendix 6.4.1). German thresholds are equal-per-capita shares of the global thresholds (Appendix 6.4.2). Underlying models as specified in EF 3.1 (Andreasi Bassi et al., 2023).

6.4.2 National thresholds

The equal-per-capita rule assigns each person the same share of the global safe operating space (Section 4.4). Let $\mathrm{PB}_{\mathrm{global}}$ denote the global threshold of an impact category and $\mathrm{Pop}_{c,t}$ and $\mathrm{Pop}_{\mathrm{world},t}$ the populations of country c and of the world in year t (United Nations, 2024). The average population share over 1995–2022,

$$\bar{\omega}_c = \frac{1}{T} \sum_{t=1995}^{2022} \frac{\mathrm{Pop}_{c,t}}{\mathrm{Pop}_{\mathrm{world},t}}, \quad T = 28,$$

gives the time-invariant national threshold $\mathrm{PB}_c = \bar{\omega}_c \cdot \mathrm{PB}_{\mathrm{global}}$.

The literature also discusses alternative allocation principles, such as equal cumulative per-capita allocation, which accounts for historical responsibility, and capability-based allocation, which accounts for differences in the ability to mitigate (Häyhä et al., 2016; Robiou du Pont et al., 2017; Lucas et al., 2020; Hjalsted et al., 2021). Both would tend to yield stricter thresholds for high-income economies than the equal-per-capita rule.

7 Decomposition and counterfactual derivations

7.1 Decomposition of impacts

We start from the definition of environmental impact and multiply by unity:

$$Impact_O = \sum_i e_{iO} = \sum_i \frac{e_{iO}}{v_{iO}} v_{iO},$$

where e_{iO} denotes the impact generated in sector i of origin O to serve German final demand, so that Impact_O equals the component $\mathbf{D}_t^O$ of Appendix 6.1.5 in that category, and v_{iO} denotes the corresponding nominal value of production. We replace the impact per euro of production for German final demand, e_{iO}/v_{iO}, with the impact per euro of total sectoral output given in the EE-MRIO tables:

$$\frac{e_{iO}}{v_{iO}} = f_{iO} \equiv \frac{q_{iO}}{x_{iO}},$$

where q_{iO} and x_{iO} denote the total impact and total output of sector i in origin O. Thus f_{iO} is the element of $\mathbf{p}_t$ for sector i (Appendix 6.3.2); for the foreign origin, it is the output-weighted average across foreign countries (Table 6).

This replacement rests on the standard IO assumption that each sector produces a homogeneous good with a single technology, whatever the destination of its output.

Substituting f_{iO} and multiplying by V_O/V_O yields equation (1) of the main text:

$$Impact_O = \underbrace{V_O}_{\text{scale}} \underbrace{\sum_i \frac{q_{iO}}{x_{iO}} \frac{v_{iO}}{V_O}}_{\text{impact intensity}} \tag{2}$$

where $V_O \equiv \sum_i v_{iO}$ is the total value of production in origin O for German final demand. The impact-intensity term, $\bar{f}_O \equiv \sum_i f_{iO}\, v_{iO}/V_O$, equals Impact_O/V_O, the impact of origin O per euro of its production for German final demand. Table 6 summarizes the notation.

7.2 Inflation and heterogeneous price levels

We deflate the value of production for German final demand separately by origin rather than at the sector-country level of the IO tables, for which price indices are unavailable for a large share of country-sector combinations. Deflation by origin still allows a meaningful comparison of the decomposition results over time. We show below under which assumption it is sufficient.

Table 6 | Notation used in the decomposition.

Variable	Definition
e_{iO}	impact generated in sector i of origin O for final demand
v_{iO}	value of production in sector i of origin O for final demand
V_O	total value of production in origin O for final demand, $\sum_i v_{iO}$
V	total value of production for final demand, $V_D + V_F$
f_{iO}	impact per euro of output in sector i of origin O
f_{iF}	output-weighted average of f_{ic} across foreign countries and regions $c \in \mathcal{F}$, $\sum_{c\in\mathcal{F}} f_{ic}\, v_{ic} / \sum_{c\in\mathcal{F}} v_{ic}$
s_{iO}	share of sector i in the value of production of origin O, v_{iO}/V_O
$\bar{f}_O$	impact intensity of origin O, $\sum_i f_{iO}\, s_{iO} = \text{Impact}_O/V_O$
S_O	share of origin O in the value of production for final demand, V_O/V; S_F is the import share
P_O	price index of origin O ($P_{O,1995} = 1$)
$\bar{P}$	price index of the combined basket, $S_D P_D + S_F P_F$

Final demand is that of the economy analyzed: Germany in the main text and each economy in turn in Appendix 9. Origin $O \in \{D, F\}$ denotes domestic production in that economy and foreign production in all other countries and regions. A subscript t denotes the year; a tilde denotes values at constant 1995 prices.

In equation (2), sector-country price changes embodied in v_{iO} and x_{iO} cancel from the impact-intensity term if the price of the output serving German final demand changes at the same rate as the price of total sectoral output. Then $v_{iO}/x_{iO} = \tilde{v}_{iO}/\tilde{x}_{iO}$, where a tilde denotes real variables.

Let P_{Ot} denote the price index of origin O in year t, with $P_{O,1995} = 1$: the German GDP deflator for domestic production and the German import price index for foreign production. For the combined basket, we use the average of the two indices weighted by the nominal shares of domestic and foreign production for German final demand, $\bar{P}_t = S_{Dt}P_{Dt} + S_{Ft}P_{Ft}$. We can then write the decomposition in terms of real variables only by multiplying by $\frac{P_{Ot}}{P_{Ot}}$:

$$Impact_{Ot} = \underbrace{\frac{V_{Ot}}{P_{Ot}}}_{\text{scale in 1995 prices}} \underbrace{\sum_i \frac{q_{iOt}}{\tilde{x}_{iOt}} \frac{\tilde{v}_{iOt}}{\frac{V_{Ot}}{P_{Ot}}}}_{\text{impact intensity in 1995 prices}} .$$

7.3 Decomposition of growth

Following Levinson (2009) and Shapiro and Walker (2018), we decompose growth relative to

a base year b:

$$\begin{aligned}\frac{Impact_t}{Impact_b} &= \frac{V_t \sum_i f_{it} s_{it}}{V_b \sum_i f_{ib} s_{ib}} \\ &= \underbrace{\frac{V_t \sum_i f_{ib} s_{ib}}{V_b \sum_i f_{ib} s_{ib}}}_{\text{Scale effect}} + \underbrace{\frac{V_b \sum_i f_{it} s_{it}}{V_b \sum_i f_{ib} s_{ib}}}_{\text{Impact-intensity effect}} + \underbrace{\frac{(V_t - V_b)(\sum_i f_{it} s_{it} - \sum_i f_{ib} s_{ib})}{V_b \sum_i f_{ib} s_{ib}}}_{\text{Interaction effect}} - 1,\end{aligned}$$

where $s_{it} = \tilde{v}_{it}/\tilde{V}_t$, and we suppress the origin subscript and the tilde for real variables. The second equation follows from adding and subtracting $V_b \sum f_{it} s_{it}$, $V_t \sum f_{ib} s_{ib}$, and $V_b \sum f_{ib} s_{ib}$, and rearranging terms. In growth rates, $1 + g_{\text{Impact}} = (1 + g_{\text{scale}}) + (1 + g_{\text{intensity}}) + g_{\text{int}} - 1 = 1 + g_{\text{scale}} + g_{\text{intensity}} + g_{\text{int}}$, where $g_{\text{int}} = g_{\text{scale}}\, g_{\text{intensity}}$.

7.4 Construction of the constant-import-share counterfactual

Summing equation (2) over the two origins yields Germany's CBA in a given impact category:

$$CBA_t = \text{Impact}_{Dt} + \text{Impact}_{Ft} = V_t \left(S_{Dt}\, \bar{f}_{Dt} + S_{Ft}\, \bar{f}_{Ft} \right),$$

where $V_t = V_{Dt} + V_{Ft}$ is the total value of production for German final demand, $S_{Ot} = V_{Ot}/V_t$ is the share of origin $O \in \{D, F\}$ in this value ($S_{Dt} + S_{Ft} = 1$), and $\bar{f}_{Ot} = \sum_i f_{iOt}\, s_{iOt}$ is the output-weighted impact intensity of origin O (Appendix 7.1). To isolate the contribution of the shift from domestic to foreign production, we hold the origin shares S_{Ot} at their 1995 values while letting the total value of production and the within-origin impact intensities evolve as observed:

$$CBA_t^{\text{cf}} = V_t \left(S_{D,1995}\, \bar{f}_{Dt} + S_{F,1995}\, \bar{f}_{Ft} \right).$$

All quantities in this counterfactual (V_t, S_{Ot}, $\bar{f}_{Ot}$) are measured in nominal terms, so that the counterfactual reallocates production values between origins at the 1995 split. We hold the nominal rather than the real share fixed because the product of nominal value and impact per current euro reproduces observed physical impacts exactly without relying on aggregate deflators, and because a constant expenditure share still lets quantities respond to relative prices.

Subtracting the counterfactual from the observed series and using $S_{Dt} + S_{Ft} = 1$ gives

$$\begin{aligned}CBA_t - CBA_t^{\text{cf}} &= V_t \Big[\big(S_{Dt} - S_{D,1995}\big) \bar{f}_{Dt} + \big(S_{Ft} - S_{F,1995}\big) \bar{f}_{Ft} \Big] \\ &= V_t \big(S_{Ft} - S_{F,1995}\big) \big(\bar{f}_{Ft} - \bar{f}_{Dt}\big).\end{aligned}$$

The difference is the product of the total value of production, the change in the import share since 1995, and the foreign–domestic gap in impact intensity. Because imports embody more impact per current euro than domestic production in every year and category ($\bar{f}_{Ft} > \bar{f}_{Dt}$; Table 10), the observed CBA exceeds its counterfactual whenever the nominal import share exceeds its 1995 level ($S_{Ft} > S_{F,1995}$).

Three qualifications apply. First, the counterfactual fixes only the aggregate import share, not the sectoral composition within either origin; both series still contain within-origin shifts in sectoral composition. Second, the value withdrawn from imports is absorbed by domestic production in proportion to its observed sectoral mix and at its observed intensity $\bar{f}_{Dt}$: a reduction in steel imports, for instance, raises domestic production across the whole domestic basket rather than domestic steel one-for-one. Third, because the counterfactual reallocates values rather than physical output, its implied quantities need not match observed final-demand quantities.

8 Robustness to incomplete stressor characterization

Our impact estimates use only those EXIOBASE stressors for which EF 3.1 provides CFs; the remaining stressors are assigned a factor of zero. Since their EF 3.1 factor is zero—most plausibly because they are not relevant to the category—they are unlikely to matter much. Nevertheless, omitting relevant stressors could in principle distort the geographic decomposition (Fig. 2), on which two findings rest: that foreign production accounts for a larger share of impacts than domestic production (a level result), and that the foreign–domestic gap widens over time (a trend result). We assess each in turn.

We assign potential uncharacterized stressors to a category by extension: any stressor from the EXIOBASE extension underlying the category's characterized impacts is treated as a candidate, even if its EF 3.1 factor is zero. This is intentionally generous—many such stressors are unlikely to be relevant to the category—so it biases the robustness check toward finding threats rather than dismissing them. Column (1) of Table 7 lists the number of uncharacterized stressors by impact category. The concern is confined to the five categories that contain uncharacterized stressors; land and water use contain none and are therefore unaffected. All five affected categories use global, origin-invariant CFs (Table 4); we therefore assume that omitted stressors would carry the same CF in domestic and foreign production.

Level: environmental impacts embodied in imports exceed impacts from domestic production We first ask whether omitted stressors could overturn the finding that imports

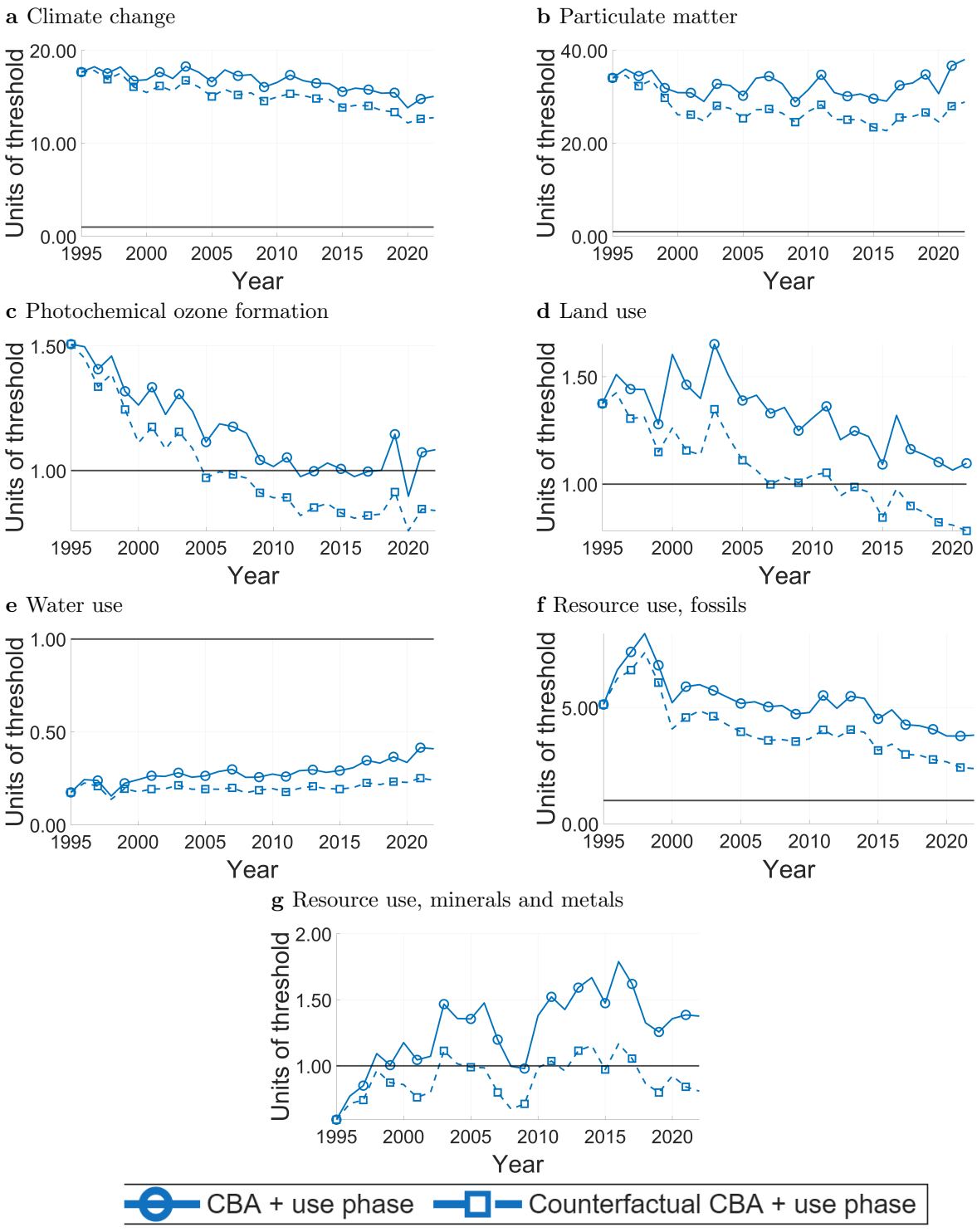


Fig. 5 | Observed and counterfactual impacts with the import share held at its 1995 level. Solid lines: observed CBA plus use-phase impacts. Dashed lines: counterfactual CBA plus observed use-phase impacts. The counterfactual holds the nominal share of imports in the value of production for German final demand at its 1995 level, while the total value of production and within-origin impact intensities evolve as observed (Appendix 7.4). The gap between the lines is the contribution of the rising import share.

dominate the footprint. Define the foreign share of a stressor as

$$\sigma^F = \frac{D^F}{D^D + D^F},$$

where D^D and D^F are its domestic and foreign components (Appendix 6.1.5). Including an omitted stressor would raise the domestic impact relative to the foreign one whenever $\sigma^F < 0.5$. Note that this is only a necessary condition for an uncharacterized stressor to reverse the gap. In the end, a gap reversal depends on the unknown CF and the joint contribution of the set of uncharacterized stressors. A joint assessment is infeasible due to a lack of a joint unit.

Table 7, Column (2), reports the time average share of uncharacterized stressors meeting this condition. It ranges from about 4% to 17% and lies below 10% in three of the five categories, so the large majority of omitted stressors would, if anything, reinforce rather than threaten the result.

Trend: the foreign–domestic gap over time We next ask whether omitted stressors could reverse the finding that domestic impacts improve while impacts embodied in imports offset these improvements in the aggregate, that is, a widening of the gap between foreign and domestic impacts.

Define

$$\mathrm{Gap}_t = D_t^F - D_t^D,$$

where D_t^F and D_t^D are the foreign and domestic components of the CBA (Appendix 6.1.5). Let $\mathrm{Gap}^{ch} = D^{F,ch} - D^{D,ch}$ denote the gap in characterized impacts and $\mathrm{Gap}^{un} = D^{F,un} - D^{D,un}$ the gap in the inventory of an omitted stressor. Including that stressor with factor CF_{un} gives the augmented gap

$$\mathrm{Gap}^{incl} = (D^{F,ch} + CF_{un} D^{F,un}) - (D^{D,ch} + CF_{un} D^{D,un}) = \mathrm{Gap}^{ch} + CF_{un}\,\mathrm{Gap}^{un}.$$

The characterized gap widens in most categories, that is, $0 < \mathrm{Gap}_{t+1}^{ch} - \mathrm{Gap}_t^{ch}$. A necessary condition for the reversal of this finding when adding an uncharacterized stressor is that

$$0 > \left(\mathrm{Gap}_{t+1}^{ch} - \mathrm{Gap}_t^{ch}\right) + CF_{un}\left(\mathrm{Gap}_{t+1}^{un} - \mathrm{Gap}_t^{un}\right). \tag{3}$$

Since the first term in brackets and CF_{un} are positive, a necessary condition is that

$$\mathrm{Gap}_{t+1}^{un} - \mathrm{Gap}_t^{un} < 0. \tag{4}$$

The share of uncharacterized stressors meeting this condition amounts to 35–40% across

categories (Table 7, Column (3)).

Within the set of uncharacterized stressors for which condition (4) holds we derive the minimum contribution of an uncharacterized stressor using equation (3), that is, a lower bound on CF_{un} so that leaving out the stressor biases our results in a relevant way:

$$CF_{un} > -\frac{\text{Gap}^{ch}_{t+1} - \text{Gap}^{ch}_{t}}{\text{Gap}^{un}_{t+1} - \text{Gap}^{un}_{t}} > 0.$$

We compare for each stressor the minimum CF_{un} with the maximum CF in the set of characterized stressors within an impact category. We report the share of uncharacterized stressors for which the necessary CF lies below the maximum CF observed in the data. These are uncharacterized stressors that may reverse our result of a widening gap under a plausibly high CF.

Column (4) of Table 7 shows the results. This share is below 3% for climate change, photochemical ozone formation, and particulate matter, but reaches 13% for minerals and 37% for fossil resource use. The trend is therefore robust for the majority of categories; the fossil-resource and, to a lesser extent, mineral-resource trends are sensitive only under the extreme assumption—assigning the category's maximum factor to stressors whose EF 3.1 factor is in fact zero.

Overall, incomplete characterization is unlikely to overturn either the level or the trend of the foreign–domestic gap, with the strongest reservation attaching to the fossil-resource trend under the most conservative possible factor assumption. Again, we cannot assess the joint importance of left-out stressors due to a lack of a joint unit.

9 International comparison

For every country covered individually in EXIOBASE except Taiwan (Section 4.7), this appendix reports three results that test whether the mechanism holds beyond Germany: boundary transgression (Appendix 9.1); the gap between foreign and domestic impact intensity that underlies the mechanism (Appendix 9.2); and the contribution of the rising import share to impact growth, quantified with the counterfactual of Section 2.4 (Appendix 9.3). Table 2 in the main text summarizes the results for a set of comparable open, high-income economies; the tables below report the full set of countries.

The implementation follows Section 4 and leaves the framework of Appendix 6 unchanged. For each country, the domestic and foreign components, $\mathbf{D}^D_t$ and $\mathbf{D}^F_t$, are defined as in Appendix 6.1.5, with the trade mask applied to the sectors of the country in question. Each country receives its equal-per-capita share of every planetary boundary, using its average

Table 7 | Robustness to missing characterization factors.

Impact category	Uncharacterized stressors	Level threats (%)	Trend threats (%)	Trend threats at maximum CF (%)
	(1)	(2)	(3)	(4)
Climate change	395	8.0	38.5	2.8
Particulate matter	272	9.2	37.5	1.5
Photochemical ozone formation	294	4.4	39.8	0.7
Land use	0	–	–	–
Water use	0	–	–	–
Resource use, fossils	51	12.4	37.3	37.3
Resource use, minerals and metals	46	17.2	34.8	13.0

Column (1): number of EXIOBASE stressors in the category left uncharacterized; because a stressor can be a candidate in several categories, the counts do not sum to the 705 stressors in the environmental extensions. Column (2): share of these stressors, averaged over time, that could threaten the level result, defined as having a foreign share below 0.5. Column (3): share with a negative trend in the foreign–domestic gap, which could in principle reverse the widening of that gap. Column (4): share that would reverse the trend result if assigned the category's maximum observed CF, a worst-case bound on Column (3). –: not defined, because land use and water use have no uncharacterized stressors.

population share over 1995–2022 exactly as in Appendix 6.4.2. Cross-country differences in transgression therefore reflect differences in per-capita footprints, not in the allocation rule.

9.1 Boundary transgression across countries

Table 8 summarizes transgression for each country by two statistics: the number of the seven impact categories in which consumption-based impacts exceed the national threshold in at least 80% of the years 1995–2022 (a measure of breadth), and the share of all $7 \times 28 = 196$ category-year observations in which the threshold is exceeded (a measure of persistence). The two statistics capture different things—a country can breach many boundaries intermittently or few boundaries continuously—which is why the ranking by persistence in Table 8 is not monotone in breadth. Table 9 disaggregates persistence by impact category for the ten highest- and lowest-ranked countries and for Germany.

Three patterns stand out. First, transgression is concentrated among high-income economies. The upper part of the ranking is occupied almost exclusively by high-income economies: Australia exceeds all seven boundaries, and 12 further economies—Luxembourg, Belgium, Finland, Canada, the United States, Switzerland, Sweden, Norway, the Netherlands, Ireland, Denmark, and Germany—consistently exceed six. The lower part is occupied by emerging and developing economies; China, India, and Indonesia consistently exceed only two

boundaries, with 29–45% of category-years above threshold. Germany, with six boundaries and 80.6% of category-years above threshold, ranks 15th of 43, within its high-income peer group: it is a representative rather than an extreme case, which is what makes it informative for the mechanism studied in the main text.

Second, the gradient is not carried uniformly by all categories. Climate change and particulate matter are exceeded in every year by every country shown in Table 9, including the ten lowest-ranked economies: at the equal-per-capita budget, even comparatively low per-capita footprints exceed these two boundaries. The cross-country spread therefore arises in land use, fossil and mineral resource use, and photochemical ozone formation, which high-income economies exceed and most lower-income economies do not. Water use, by contrast, remains within the boundary for all countries shown except Australia (96% of years) and Luxembourg (32%); Germany's compliance in this category (Section 2.1) is thus not a German peculiarity.

Third, the breadth statistic is sensitive to the 80% cut-off for countries with categories close to it. For Germany, mineral resource use and photochemical ozone formation are exceeded in 82% of years; at a 90% cut-off, Germany's count would fall from six to four. The persistence statistic is unaffected by this choice, and the ranking is robust to it, but the exact number of "consistently exceeded" boundaries should be read with the cut-off in mind.

9.2 Foreign versus domestic impact intensity across countries

Table 10 reports, for each country and category, the share of years in which foreign production for the country's final demand carries more impact per current euro than domestic production. In 40 of the 43 economies, foreign intensity exceeds domestic intensity in more than half of all category-years, that is, the mean exceeds 50% (Column (8)). In 23 economies, foreign intensity exceeds domestic intensity in at least 90% of category-years, and in 13 of them (Austria, Belgium, Denmark, Finland, France, Germany, Hungary, Italy, Japan, Korea, Luxembourg, Switzerland, and the United Kingdom) in every category-year; the Netherlands and the United States fall short of this in a single category (fossil and mineral resource use, respectively), where the ordering holds in 27 of the 28 years (96%). The ordering fails in most category-years only in South Africa, Russia, and China (means of 30–46%), with India close behind (52%): large middle-income or resource-rich economies whose domestic production is itself comparatively impact-intensive.

We compare intensities at current prices in the main analysis because deflation would not remove the price-level differences between countries, which both measures reflect, but would only rescale the foreign–domestic intensity ratio by the change in relative prices since an arbitrary base year. At current prices, however, the comparison also reflects how the prices

Table 8 | Boundary transgression by country, 1995–2022.

Rank	Country	Boundaries consistently exceeded (of 7)	Category-years above threshold (%)	Rank	Country	Boundaries consistently exceeded (of 7)	Category-years above threshold (%)
1	AU	7	99.5	23	ES	4	74.0
2	LU	6	89.8	24	IT	4	70.9
3	BE	6	85.7	25	CY	4	67.6
4	FI	6	85.7	26	PT	4	67.3
5	CA	6	85.7	27	MX	4	66.2
6	US	6	85.7	28	LT	4	62.2
7	CH	6	85.2	29	LV	4	62.2
8	SE	6	84.7	30	BG	4	59.3
9	NO	6	84.7	31	CZ	3	57.9
10	NL	6	84.2	32	RU	4	57.8
11	IE	6	83.7	33	PL	4	57.7
12	DK	6	83.2	34	ZA	4	55.1
13	AT	5	81.6	35	BR	3	54.6
14	GB	5	81.1	36	HR	3	54.3
15	**DE**	**6**	**80.6**	37	RO	3	50.8
16	GR	5	78.6	38	SK	3	50.1
17	KR	5	78.6	39	TR	3	49.0
18	JP	5	76.5	40	HU	3	45.4
19	SI	5	76.0	41	CN	2	45.4
20	MT	4	75.9	42	IN	2	29.1
21	FR	4	75.0	43	ID	2	28.6
22	EE	5	74.5				

Boundaries consistently exceeded: number of the seven impact categories in which the country's consumption-based impacts exceed its national threshold in at least 80% of the years 1995–2022. Category-years above threshold: share of all $7 \times 28 = 196$ category-year observations in which the threshold is exceeded. Countries are ranked by this share.

of domestic output and of imports have evolved: if domestic prices rise faster than import prices, domestic intensity per current euro falls relative to foreign intensity even when physical intensities do not change. To check that our results are not driven by such differences in price developments, we repeat the comparison with series deflated to constant 1995 prices, as in the trend analysis for Germany (Table 11).

For each country, domestic production is deflated with its GDP deflator from the World Bank and foreign production with its deflator of imports of goods and services from the European Commission's AMECO database (EUR/ECU basis). Import price deflators are available for 36 of the 43 economies, which the check therefore covers.

The deflated comparison confirms the main results: Germany and the comparable open, high-income economies (Table 2) are virtually unaffected, and the shares fall markedly only in a few middle-income economies, most strongly in Turkey, Bulgaria, and Mexico.

Table 9 | Boundary transgression by impact category: highest- and lowest-ranked countries, 1995–2022.

Rank	Country	Climate change (%)	Particulate matter (%)	Photo-chemical ozone formation (%)	Land use (%)	Water use (%)	Resource use, fossils (%)	Resource use, minerals and metals (%)
Top 10								
1	AU	100	100	100	100	96	100	100
2	LU	100	100	100	100	32	100	96
3	BE	100	100	100	100	0	100	100
4	FI	100	100	100	100	0	100	100
5	CA	100	100	100	100	0	100	100
6	US	100	100	100	100	0	100	100
7	CH	100	100	100	100	0	100	96
8	SE	100	100	93	100	0	100	100
9	NO	100	100	100	96	0	100	96
10	NL	100	100	100	100	0	100	89
Germany								
15	**DE**	**100**	**100**	**82**	**100**	**0**	**100**	**82**
Bottom 10								
34	ZA	100	100	0	0	0	96	89
35	BR	100	100	0	100	0	46	36
36	HR	100	100	7	52	0	100	21
37	RO	100	100	0	52	0	100	4
38	SK	100	100	0	26	0	96	29
39	TR	100	100	0	0	0	96	46
40	HU	100	100	11	4	0	100	4
41	CN	100	100	0	4	0	68	46
42	IN	100	100	0	0	0	4	0
43	ID	100	100	0	0	0	0	0

Share of the years 1995–2022 in which the country's consumption-based impacts exceed its national threshold in each category. Countries are ranked by their share of category-years above threshold (Table 8); the table shows the ten highest- and ten lowest-ranked countries and Germany.

9.3 Contribution of the import share across countries

Table 12 reports, for each country and category, the percentage-point difference between actual cumulative impact growth over 1995–2022 and growth under the constant-import-share counterfactual. As for Germany, the counterfactual holds the share of foreign production in the value of production for that country's final demand at its 1995 value (Appendix 7.4). A positive entry means that the observed change in the domestic–foreign production mix raised impact growth.

Among the comparable open, high-income economies of Table 2—Belgium, France, Germany, Italy, Japan, Korea, the Netherlands, Spain, the United Kingdom, and the United

States—the effect is positive in every one of the seven categories for every country: the rising import share raised cumulative impact growth in all 70 country-category cells. Across all 41 countries other than Ireland and Luxembourg, the mean effect across categories is positive in 36 and the effect is positive in every category in 18. As for Germany, effects are on average largest for water use and mineral resource use and smallest for land use, climate change, and photochemical ozone formation. Spain is an exception for water use, where its effect is the smallest of all seven categories. Germany's mean effect of 42.8 percentage points exceeds that of every peer (16–35 percentage points), which suggests that the German case overstates the typical magnitude of the composition channel among comparable economies while being representative of its sign.

The mean effect is negative in the remaining five economies, China, Mexico, Canada, Russia, and Norway, most of which have a large domestic resource base. In China and Russia, foreign production is not more impact-intensive than domestic production in most category-years (Table 10).

Table 10 | Foreign versus domestic impact intensity by country, current prices.

Country	Climate change (%) (1)	Particulate matter (%) (2)	Photo-chemical ozone formation (%) (3)	Land use (%) (4)	Water use (%) (5)	Resource use, fossils (%) (6)	Resource use, minerals and metals (%) (7)	Mean (%) (8)
LU	100	100	100	100	100	100	100	100
MT	100	68	93	100	100	100	100	94
IE	100	96	96	81	100	100	100	96
SK	93	86	100	89	100	100	96	95
BE	100	100	100	100	100	100	100	100
EE	86	68	100	48	100	100	100	86
HU	100	100	100	100	100	100	100	100
NL	100	100	100	100	100	96	100	99
CY	100	71	50	100	100	100	100	89
SI	100	100	71	100	100	100	100	96
LT	89	96	93	19	100	100	100	85
CZ	79	100	96	100	100	96	100	96
LV	93	93	93	33	100	100	100	87
CH	100	100	100	100	100	100	100	100
BG	57	61	71	7	100	100	14	59
AT	100	100	100	100	100	100	100	100
DK	100	100	100	100	100	100	100	100
SE	100	100	100	100	100	100	11	87
HR	100	100	100	96	100	100	96	99
PL	25	89	86	96	100	96	11	72
FI	100	100	100	100	100	100	100	100
KR	100	100	100	100	100	100	100	100
PT	100	100	100	100	96	100	93	98
NO	100	100	100	100	100	64	100	95
RO	64	68	64	4	100	96	89	69
CA	100	100	100	78	100	96	0	82
DE	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
MX	100	89	86	44	86	100	0	72
GR	100	64	54	93	0	100	96	72
FR	100	100	100	100	100	100	100	100
ES	100	100	100	100	25	100	93	88
GB	100	100	100	100	100	100	100	100
ID	54	100	71	0	64	100	39	61
RU	0	50	25	59	100	21	11	38
IT	100	100	100	100	100	100	100	100
ZA	0	0	0	30	86	93	4	30
TR	100	96	100	100	25	100	96	88
CN	29	14	50	93	7	79	54	46
AU	93	100	100	48	61	89	4	71
IN	0	32	71	63	0	96	100	52
JP	100	100	100	100	100	100	100	100
US	100	100	100	100	100	100	96	99
BR	43	61	93	0	100	100	89	69

Share of the years 1995–2022 in which the impact intensity of foreign production for the country's final demand exceeds that of domestic production, measured at current prices. Column (8) reports the unweighted mean across the seven impact categories, which equals the share of all category-years. Countries are listed in descending order of trade openness (World Bank).

Table 11 | Foreign versus domestic impact intensity by country, constant 1995 prices.

Country	Climate change (%) (1)	Particulate matter (%) (2)	Photo-chemical ozone formation (%) (3)	Land use (%) (4)	Water use (%) (5)	Resource use, fossils (%) (6)	Resource use, minerals and metals (%) (7)	Mean (%) (8)
LU	100	100	100	100	100	100	100	100
MT	100	79	96	100	100	100	100	96
IE	100	93	93	63	100	100	100	93
SK	93	82	100	89	100	100	96	94
BE	100	100	100	100	100	100	100	100
EE	32	54	71	26	100	100	100	69
HU	54	75	82	37	100	100	100	78
NL	100	100	100	100	100	96	100	99
CY	100	71	50	100	100	100	100	89
SI	100	100	68	100	100	100	100	95
LT	89	96	89	15	100	100	100	84
CZ	64	96	79	100	100	96	100	91
LV	93	93	93	22	100	100	100	86
CH	100	100	100	100	100	100	100	100
BG	0	0	4	0	7	7	0	3
AT	100	100	100	100	100	100	100	100
DK	100	100	100	100	100	100	100	100
SE	100	100	100	100	100	100	7	87
HR	100	100	100	85	100	100	89	96
PL	0	68	57	52	100	96	4	54
FI	100	100	100	100	100	100	100	100
KR	100	100	100	100	100	100	100	100
PT	100	100	100	100	93	100	86	97
NO	100	93	89	100	100	50	100	90
RO	–	–	–	–	–	–	–	–
CA	100	100	100	44	100	96	0	77
DE	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
MX	29	14	0	19	18	100	0	26
GR	93	46	50	89	0	100	93	67
FR	100	100	100	100	100	100	100	100
ES	100	100	100	100	7	100	93	86
GB	100	100	100	100	100	100	100	100
ID	–	–	–	–	–	–	–	–
RU	–	–	–	–	–	–	–	–
IT	100	100	100	100	100	100	100	100
ZA	–	–	–	–	–	–	–	–
TR	7	0	4	22	0	21	7	9
CN	–	–	–	–	–	–	–	–
AU	86	100	100	11	7	71	0	54
IN	–	–	–	–	–	–	–	–
JP	100	100	100	100	100	100	100	100
US	100	100	100	100	100	100	96	99
BR	–	–	–	–	–	–	–	–

As Table 10, but with domestic production deflated by the country's GDP deflator (World Bank) and foreign production by its deflator of imports of goods and services (AMECO, EU-R/ECU basis). –: no import price deflator available.

Table 12 | Actual versus counterfactual growth in environmental impacts, by country.

Country	Climate change (pp) (1)	Particulate matter (pp) (2)	Photo-chemical ozone formation (pp) (3)	Land use (pp) (4)	Water use (pp) (5)	Resource use, fossils (pp) (6)	Resource use, minerals and metals (pp) (7)	Mean (pp) (8)
AT	8.4	12.0	10.2	4.8	38.8	24.5	34.0	19.0
BE	20.1	32.2	19.4	16.6	55.6	36.6	49.0	32.8
BG	3.0	8.1	7.0	−0.3	70.6	12.2	−10.0	12.9
CY	0.6	2.6	1.6	−0.4	6.8	3.2	10.0	3.5
CZ	4.7	7.9	6.8	4.8	51.0	13.1	125.7	30.6
DE	**12.9**	**26.9**	**16.2**	**22.9**	**96.8**	**28.0**	**95.7**	**42.8**
DK	13.9	17.6	11.9	12.8	21.4	24.1	59.5	23.0
EE	−1.7	4.2	6.0	−20.9	98.1	101.9	80.4	38.3
ES	17.9	23.7	13.4	8.4	3.5	91.1	32.7	27.2
FI	3.1	8.6	5.2	6.1	23.8	18.1	2.9	9.7
FR	12.4	15.7	11.4	6.4	48.5	18.5	52.0	23.6
GR	4.7	0.6	−0.3	−5.7	−5.5	39.7	−8.0	3.6
HR	9.1	8.9	4.2	−0.5	36.2	19.3	62.2	19.9
HU	12.2	12.5	12.2	6.5	79.8	35.1	135.8	42.0
IE	98.2	63.1	50.2	50.5	611.0	246.7	358.4	211.2
IT	8.6	13.8	7.4	6.4	16.0	23.5	39.3	16.4
LT	5.5	14.4	3.0	−3.6	61.8	224.3	450.9	108.0
LU	97.7	156.3	85.4	166.6	469.5	120.6	124.5	174.4
LV	0.9	1.2	1.2	−0.2	17.1	34.7	27.4	11.8
MT	4.4	4.5	3.0	0.4	9.4	5.9	9.3	5.3
NL	14.4	23.5	16.4	15.4	67.9	31.0	78.2	35.3
PL	−0.9	8.4	5.9	5.6	174.5	10.4	−2.7	28.7
PT	5.8	8.3	4.8	5.8	6.7	20.9	−9.7	6.1
RO	3.2	7.3	6.8	−2.3	81.8	8.2	32.0	19.6
SE	8.6	14.6	8.7	5.5	33.3	33.0	−24.1	11.4
SI	8.3	8.4	−46.5	1.8	52.4	11.8	36.8	10.4
SK	16.8	12.4	19.2	6.0	252.0	234.0	103.7	92.0
GB	7.8	14.7	6.7	5.1	46.3	14.9	43.6	19.9
US	8.7	27.5	7.3	13.9	28.8	18.9	32.4	19.6
JP	14.8	22.9	10.2	16.0	46.2	43.2	66.7	31.4
CN	−9.5	−2.6	−10.2	−63.6	−2.0	−66.5	−59.2	−30.5
CA	−1.2	−2.8	−1.0	−0.3	−3.2	−1.6	0.9	−1.3
KR	9.8	17.0	9.0	12.8	30.0	45.1	48.5	24.6
BR	−6.8	−0.8	−1.2	−7.1	67.2	36.3	7.9	13.6
IN	−13.4	−0.3	6.9	14.1	−37.6	81.8	277.4	47.0
MX	1.1	−0.4	−0.4	−2.5	−27.4	34.4	−22.8	−2.6
RU	0.6	−0.2	0.2	−0.0	−10.0	0.3	0.3	−1.3
AU	2.0	4.4	2.4	−0.0	0.0	1.4	−0.5	1.4
CH	15.0	8.8	4.9	4.3	20.5	15.0	32.5	14.4
TR	5.4	4.1	4.7	28.4	−0.3	77.7	9.5	18.5
NO	−0.0	−0.0	−0.0	−2.3	−0.1	0.0	−0.1	−0.4
ID	0.1	1.0	0.5	−0.5	1.0	6.0	0.4	1.2
ZA	−2.8	−2.6	−2.0	−1.4	11.0	6.8	0.0	1.3

Percentage-point difference between actual cumulative growth of consumption-based impacts over 1995–2022 and growth under a counterfactual that holds the import share at its 1995 level, measured at current prices (Appendix 7.4). Positive values indicate that the observed change in the domestic–foreign sourcing mix raised impact growth. Column (8) reports the unweighted mean across the seven impact categories.